

OAKWOOD BUILDING AT KALYAN

OAKWOOD BUILDING -Kalyan			
SCHEDULE OF QUANTITIES			
Item Nos.	Description	Unit	Qty.
A.	SANITARY FIXTURES & FITTINGS (INSTALLATION)		
	Note:		
	i. Detail of Sanitary fixture are for the information of the Contractor, however model / makes of all sanitary fixture shall be selected by Architect / Interior designer / client and the same shall be binding for execution .		
	ii. All sanitary fixture and accessories / CP fittings shall be issued free of cost by the employer alongwith standard accessories. The contractor shall supply all fixing / supporting / accessories, devices, nuts, bolts, screws, hangers C.P. connector, elbows, extn. PSC & S.S. flanges as required and install the fixture / fitting in a neat workman like manner.		
	iii. No additional all fixing cost shall be paid for change in type of sanitary fixture or fitting.		
	iv. Provision of extension piece for final connection of CP fitting shall be supplied and installed by the contractor accordingly (as required).		
	v. Flaring should be done to C.P. Connectors before fixing it to Heaters & other fixtures.		
	vi. CP fittings and sanitary ware fittings should be fixed after acid washing is complete. All fittings shall be fixed with Teflon tape & white zinc (Safeda) only and not Hold tite or Super tite. Before fixing of Cp fittings, A) Cement grouting should be done around concealed elbows. (White cement in contractors scope) B) Cloths/duster should be used to fix CP fittings to avoid scratches/ damage etc. C) Water testing / cleaning shall be done of lines		

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Item Nos.	Description	Unit	Qty.
1	Fixing, testing and commissioning of white vitreous China wall mounting type European water closet with C.P. bolts, nuts, C.I. chair or other hanging arrangements, bakelite seat & cover with C.P. hinges & rubber buffers, C.P. brass screw, washer with all accessories. Including cutting & making good the walls, floors, slab wherever required.	Nos.	0.00
2	Fixing, testing and commissioning of Concealed Cistern for wall hung water closet with supporting arrangements, with proper line level and required civil works, complete with all accessories. Including cutting & making good the walls & floors wherever required.	Nos.	0.00
3	Fixing, testing and commissioning of white vitreous china floor mounting type European water closet with built-in exposed cistern of capacity 3 & 6 litres with C.P. bolts, nuts, supporting arrangements, bakelite seat & cover with C.P. hinges & rubber buffers, complete with all accessories. Including cutting & making good the walls & floors wherever required.	Nos.	396.00
4	Fixing, testing and commissioning of white vitreous china floor mounting type EWC or Anglo Indian water closet without cistern with C.P. bolts, nuts, supporting arrangements, bakelite seat & cover with C.P. hinges & rubber buffers, complete with all accessories. Including cutting & making good the walls & floors wherever required. (For Gr. Toilet)	Nos.	1.00
5	Fixing, testing and commissioning of exposed plastic cistern for WC, complete with all necessary fittings, flush pipes & accessories including all supports and making good. cistern dual flush, outlet from tank to w.c. (For Gr. Toilet)	Nos.	1.00
6	Fixing, testing & commissioning of Exposed Single lever wall mixer assembly with 150 mm long bend pipe OR hand shower with flexible pipe / shower rail and wall flanges complete as a full set as required.	Nos.	298.00

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Item Nos.	Description	Unit	Qty.
7	Fixing, testing & commissioning of Concealed Single lever wall divertor assembly with hand shower with flexible pipe, spout & hook for hand shower including O.H. Shower with arm C.P. wall flange complete as required.	Nos.	0.00

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Item Nos.	Description	Unit	Qty.
8	Fixing, testing and commissioning of exposed single lever sink tap wall or table mounted with flange complete with all necessary fittings, and making good.	Nos.	198.00
9	Fixing, testing & commissioning of ABS Rinsing spray / health faucet / Jet Plate etc. with regulator of approved quality with plastic flexible pipe 1.0 m long, wall hooked complete as required.	Nos.	397.00
10	Fixing, testing and commissioning of vitreous china below counter / above counter wash basin / semi counter wash basin with CP brass waste assembly, CP cast brass bottle trap with extension piece, wall flanges and rubber adapters for waste connection complete including CI / MS brackets duly painted, cutting and making good the walls floors wherever required	Nos.	0.00
11	Fixing, testing and commissioning of vitreous china wall hung flat back wash basin with C.I./M.S. brackets duly painted, C.P. brass waste, 32 mm diameter PP / C.P. cast brass bottle trap with extension piece and wall flange complete, including cutting and making good the walls/floors wherever required.	Nos.	397.00
12	Fixing, testing and commissioning of C.P. brass pillar cock "including providing & fixing connector" for pillar cock complete as required.	Nos.	397.00
13	Fixing, testing & commissioning of instant / storage type water heater of approved shade with copper / SS container, glasswool insulation, stove enamelled M.S jacket, thermostatically controlled inner heater with pilot neon lamps, suitable for working pressure of 5/ 8 kg/sqcm with suitable KW heater coil, 240 volts, single phase supply complete as required. <u>Including providing and fixing of g.i. brackets, anchor fastener, g.i. nut-bolt-washer etc. complete. (Bathroom heater fixing above false ceiling)</u>		
13.1	Capacity 3 litres (3KW)	Nos.	0.00
13.2	Capacity 3 litres (4.5KW)	Nos.	298.00

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Item Nos.	Description	Unit	Qty.
14	Fixing, testing and commissioning of C.P. brass Bib cock with C.P. wall flange , Washer etc. complete as required. (for Refuge flat / S. Toilet etc.)	Nos.	2.00
15	Fixing, testing and commissioning of C.P. brass angle valve with C.P. wall flange , Nut and Washer etc. complete as required. (for heater inlet, health faucet, exposed flush tank inlet, W/M inlet, Purifier inlet etc.) W/M & Purifier inlet Angle cock with coupling and plug.	Nos.	1488.00
16	Providing, Fixing, testing and commissioning of S.S. 5" dia. round jali for floor traps etc. complete as required. (For refuge flats) 5" Round (Glossy finish) 2PC	Nos.	3.00
17	Providing, Fixing, testing and commissioning of 5" x 5" cockroach trap S.S. Jali for floor traps etc. complete as required. (for Kitchen sinks, pantry etc.) 5" x 5" Square (Glossy finish), 18 gauge, Cockroach Trap	Nos.	198.00
18	Providing, Fixing, testing and commissioning of 5" x 5" Square (Matt finish), 18 gauge 2PC (For Bathrooms, balcony / deck drain)	Nos.	596.00
	Note:- All accessories like screws for fixing sanitary ware or brackets, anchor fasteners, additional c.p. flanges, c.p. connectors, c.p. male threaded elbow etc. shall be supplied and fixed by contractor to complete the installation.		
	Scope also includes Testing & commissioning of all fixtures (CP/Sanitary) installed in sample flat, including maint. of same.		
	TOTAL : "A" CARRIED TO SUMMARY		

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Item Nos.	Description	Unit	Qty.
B.	COLD & HOT WATER SUPPLY (INTERNAL)		
1	Concealed Piping (Hot & Cold water supply in Toilets and Kitchens)		
1.1	Providing, fixing, jointing and testing in position CPVC (Chlorinated Poly Vinyl Chloride) water supply pipes with fittings as per CTS SDR 11 (operating pressure - 7 Bar @ 82 Deg C and 28 Bar @ 23 Deg C).		
	Pipes shall be joined using solvent welded CPVC fittings i.e. Tees, Elbows, Couplers, Unions, Reducers, bushings etc. including transition fittings i.e. Brass Adaptors / Elbows (both Male & Female threaded) conforming to ASTM D-2846. ASTM F441 with only CPVC solvent cement conforming to ASTM F-493.		
	Cost shall be inclusive of (a) Making chasing in wall for the pipe, making good the same after piping and pressure testing by using 1:1 Geru mixed cement mortar <u>including all Civil materials</u> . or (b) Fixing/supporting the pipes (& fittings) at wall/ceiling level supported by split clamp with rubber lining type hangers etc, as per specification. PVC pipe sleeves of suitable higher size shall be provided wherever the pipes are crossing the walls / floors slab.		
	(c) Pressure testing of all concealed water supply pipes as per requirement (5 kg/cm2 for Gravity system & 10 kg/cm2 for Hydropneumatic system) (d) Curing of chasing finishing works for 7 days.		
	For hot & Cold water piping in toilets.		
	Nitrile rubber insulation for exposed Hot water pipes in ceiling or in shaft or on terrace. (15 to 25mm - 9mm thk. / 32 to 80mm = 13mm thk.)		
1.1.1	15 mm dia (CPVC pipes with fittings as per CTS SDR 11)	RM	4500.00
1.1.2	20 mm dia (CPVC pipes with fittings as per CTS SDR 11)	RM	80.00

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Item Nos.	Description	Unit	Qty.
2	G.I. PIPING		
2.1	Providing, fixing, jointing and testing in position the following heavy class (Class-C) G.I. pipes conforming to IS:1239 cut to required lengths including all necessary fittings (All fittings shall confirm to IS:1879 - part 1 to 10) and specials such as bends, tees, unions, reducers, flanges & plugs etc. Fixing at wall/ceiling level supported by galvanized clamps, hangers etc, as per specification. Threading, jointing, and making proper connections.		
	For pump delivery lines only 'C' type Fabricated Hotdip galvanised brackets with PVC coated 'U' clamps shall be used for vertical and horizontal supports as per details given below		
	Vertical support spacing (Center to Center) 'C' type Fabricated Hotdip galvanised brackets with GI 'U' clamps - shall be as follows -15mm = 1800mm / 20mm & 25mm = 2000mm / 32mm to 50mm = 2800mm / 65mm & above = 3000mm Horizontal supports spacing (Center to Center) shall be as follows - 15mm = 1250mm / 20mm & 25mm = 2000mm / 32mm to 50mm = 2200mm / 65mm & above = 2500mm. Above are manufacturer's standards, but for vertical supports use 3 brackets / floor for residential building and 4 brackets / floor for commercial building or 1 m c/c whichever is less. For horizontal supports use 1.25 m c/c for all sizes.		
	For exposed piping for TMC, Pump Delivery-Suction, Tank Interconnection etc. O.H. Tank to Podium loop lines		
2.1.1	50 mm dia (For municipal connection)	RM	20.00
2.1.2	65 mm dia (Flushing delivery)	RM	150.00
2.1.3	80 mm dia (Domestic delivery)	RM	150.00
2.1.4	100 mm dia	RM	6.00
2.1.5	150 mm dia (Domestic & Flushing Suction, Tank interconnection)	RM	12.00

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Item Nos.	Description	Unit	Qty.
2.2	Providing, fixing, jointing and testing in position the following heavy class (Class-C) G.I. pipes conforming to IS:1239 cut to required lengths with Zoloto-M or ' R ' brand GI fittings as per specifications and complete with fittings such as sockets, unions, reducers long or short bends, elbows, nipples flanges, checknuts, clamps etc.		
	Scope includes, Flanged connections with G.I. puddle sleeves provided in RCC walls with providing and fixing of required G.I. flanges, G.I. Nut-Bolt & Washers & rubber packing's.		
	Scope also includes Excavation of required width & depth, Pressure testing of piping and providing protection to embedded G.I. Pipes & fittings by wrapping hassion cloth over two coats of bitumen, & back filling of trench complete as required.		
	For under ground piping for TMC including all Civil materials. (excavation, backfilling quantity considered separately)		
2.2.1	50 mm dia (For municipal connection)	RM	10.00
2.2.2	80 mm dia	RM	0.00
3	Providing, fixing, jointing and testing in position the following U-PVC pressure Pipes Sch. 40 as per ASTM D 1785 and plain end (pasted) . Cut to required lengths including all necessary fittings and specials such as bends, tees, unions, reducers, plugs etc in U-PVC pressure fittings Sch. 80 as per ASTM D 1785 Jointing with solvent cement. Use Cleaner, Primer and solvent for pipe dia. 50mm and more.		
	Scope includes, Flanged connections with G.I. puddle sleeves provided in RCC walls with providing and fixing of required G.I. flanges, G.I. Nut-Bolt & Washers & rubber packing's.		
	Scope also includes Pressure testing of piping, encasing of pipes with CC M-15 & back filling of trench complete as required.		
	For under ground piping for, Irrigation, STP, Sump pumps , Tanker inlet, DCB etc.		
	including all Civil materials. (excavation, concrete, backfilling quantity considered separately)		
3.1	50 mm dia (U.G. STP, TMC line, Irrigation line etc.)	RM	10.00

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Item Nos.	Description	Unit	Qty.
3.2	80 mm dia (U.G. Sump pump delivery, DCB, Tanker inlet, Irrigation line etc.)	RM	400.00
3.3	100 mm dia (U.G. Tanker inlet / Sump pump delivery / STP line)	RM	10.00

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Item Nos.	Description	Unit	Qty.
4	Providing, fixing, jointing and testing in position the following U-PVC pressure Pipes Sch. 40 as per ASTM D 1785 and plain end (pasted) . Cut to required lengths including all necessary fittings and specials such as bends, tees, unions, reducers, plugs etc in U-PVC pressure fittings Sch. 80 as per ASTM D 1785 Jointing with solvent cement . <u>Including G.I. Brackets with PVC coated 'U' clamps fixing at wall/ ceiling level, hangers etc. as per specification. Cutting hole in wall and making good the same with cement concrete M-15 complete as required. pvc pipe sleeve of larger diameter shall be provided wherever the pipes crossing the walls / floor slab and sealing the sleeves as per consultants requirement. Use Cleaner, Primer and solvent for pipe dia. 50mm and more.</u>		
	Brass insert male adaptor shall be used for toilet / kit. inlet to branch pipe connection (15mm & 20mm). For terrace loop branch valves, metal insert adaptors shall be used.		
	For suspended piping (other than loop) GI split clamp with rubber lining with rod/nut/washer etc. shall be used as per required spacing. <u>Every 5 mtr. Interval G.I. bracket {including painting} with pvc coated clamps shall be used.</u>		
	Scope also includes miscellaneous works like – U.G. & O.H. tank overflow, drain {Fire/Flush/Domestic} , 25 dia. loop air piping with supports, mosquito nets etc.		
	for vertical supports use 3 brackets / floor for residential building and 4 brackets / floor for commercial building or 1 m c/c whichever is less. Additional supports shall be provided for water supply branch pipes in shaft, which are more than 500mm long. (2 supports, if branch is more than 1000mm long)		
	For vertical lines, Terrace loop & branch pipes, 'C' type Fabricated Hotdip galvanised brackets with PVC coated 'U' clamps shall be used.		
	Horizontal supports spacing (Center to Center) shall be as follows - 15mm and 20mm = 500mm / 25mm = 600mm / 32mm = 700mm / 40mm = 800mm / 50mm = 900mm and 65mm and bigger dia. = 1000mm.		

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Item Nos.	Description	Unit	Qty.
	(For External water supply downtake / Uptake piping in shaft, staircase servant toilet open piping, Terrace loop, STP line, DCB line, tanker line, sump pump lines etc.)		

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Item Nos.	Description	Unit	Qty.
4.1	15 mm dia	RM	1470.00
4.2	20 mm dia	RM	675.00
4.3	25 mm dia	RM	1400.00
4.4	32 mm dia	RM	1350.00
4.5	40 mm dia	RM	800.00
4.6	50 mm dia (Terrace loop, tank overflow, tank drain, STP line etc.)	RM	663.00
4.7	65 mm dia (Terrace loop)	RM	210.00
4.8	80 mm dia (Exposed Sump pump delivery, Terrace loop, DCB, Tanker inlet, Irrigation line, STP line etc.)	RM	472.00
4.9	100 mm dia (Exposed Sump pump delivery, Terrace loop, Tanker inlet, STP line etc.)	RM	264.00
5	Providing, fixing, jointing and testing in position the following U-PVC pressure Pipes Sch. 80 as per ASTM D 1785 and plain end (pasted). Cut to required lengths including all necessary fittings and specials such as bends, tees, unions, reducers, plugs etc in U-PVC pressure fittings Sch. 80 as per ASTM D 1785 Jointing with solvent cement. Including G.I. Brackets with PVC coated 'U' clamps fixing at wall/ ceiling level, hangers etc. as per specification. Cutting hole in wall and making good the same with cement concrete M-15 complete as required. pvc pipe sleeve of larger diameter shall be provided wherever the pipes crossing the walls / floor slab and sealing the sleeves as per consultants requirement. Use Cleaner, Primer and solvent for pipe dia. 50mm and more.		
	Brass insert male adaptor shall be used for toilet / kit. inlet to branch pipe connection (15mm & 20mm). For terrace loop branch valves, metal insert adaptors shall be used.		

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Item Nos.	Description	Unit	Qty.
	For suspended piping (other than loop) GI split clamp with rubber lining with rod/nut/washer etc. shall be used as per required spacing. <u>Every 5 mtr. Interval G.I. bracket {including painting} with pvc coated clamps shall be used.</u>		

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Item Nos.	Description	Unit	Qty.
	Scope also includes miscellaneous works like – U.G. & O.H. tank overflow, drain {Fire/Flush/Domestic} , 25 dia. loop air piping with supports, mosquito nets etc.		
	for vertical supports use 3 brackets / floor for residential building and 4 brackets / floor for commercial building or 1 m c/c whichever is less. Additional supports shall be provided for water supply branch pipes in shaft, which are more than 500mm long. (2 supports, if branch is more than 1000mm long)		
	For vertical lines & branch pipes, 'C' type Fabricated Hotdip galvanised brackets with PVC coated 'U' clamps shall be used.		
	Horizontal supports spacing (Center to Center) shall be as follows - 15mm and 20mm = 500mm / 25mm = 600mm / 32mm = 700mm / 40mm = 800mm / 50mm = 900mm and 65mm and bigger dia. = 1000mm.		
	(For External water supply downtake piping in shaft for 50mm & bigger dia.)		
5.1	50 mm dia (sch. 80)	RM	0.00
5.2	65 mm dia (sch. 80)	RM	0.00
5.3	80 mm dia (sch. 80)	RM	0.00
5.4	100 mm dia (sch. 80)	RM	0.00
6	Providing & fixing full way lever operated forged brass ball valve of brass body with forged brass hard chromeplated steel ball tested to a pressure not less than 20 Kg / sqcm with threaded / flanged joints complete with nuts, bolts, gaskets, washers etc.		
	For all Water supply branch on terrace / podium level and in shafts, Terrace loop, Hydropneumatic loop, U.G. & O.H. tank drain {Fire/Flush/Domestic}, pump priming, before & after PRV, below air valve, Municipal etc.		
	PVC (male / female) adaptors can be used for Ball Valves of water supply downtake branch pipes in toilet & kitchen shaft. Union shall be provided near all ball valves.		

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Item Nos.	Description	Unit	Qty.
	Brass Insert (male / female) adaptors shall be used for Ball Valves of before & after PRV. Union shall be provided near all ball valves.		

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Item Nos.	Description	Unit	Qty.
6.1	15 mm dia	Nos.	960.00
6.2	20 mm dia	Nos.	50.00
6.3	25 mm dia	Nos.	228.00
6.4	32 mm dia	Nos.	174.00
6.5	40 mm dia	Nos.	18.00
6.6	50 mm dia	Nos.	31.00
6.7	65 mm dia (For loop)	Nos.	2.00
6.8	80 mm dia (For loop)	Nos.	2.00
7	Providing & fixing butterfly valve (upto 150mm dia with hand lever operation & above with gear box operation). Tested to a pressure not less than as mentioned below, Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers.		
	For tank interconnection, pump suction / delivery, STP, bore well, Tanker inlet, Sump Pump, RWH lines etc.		
7.1	65 mm dia PN-25 gear operated.	Nos.	0.00
7.2	65 mm dia (Main Flushing pump) PN-16	Nos.	5.00
7.3	65 mm dia (OHT) PN-10	Nos.	0.00
7.4	100 mm dia (Main Domestic, Flushing pump) PN-16	Nos.	0.00
7.5	80 mm dia (DCB, Sump Pump, Irrigation line, Tanker line, STP line, Tanker line, OHT) PN-10.	Nos.	17.00
7.6	80 mm dia (Main Domestic pump) PN-16.	Nos.	5.00
7.7	100 mm dia (Suction header, Sump Pump, Irrigation pump, Loop, OHT, Tanker line) PN-10.	Nos.	9.00

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Item Nos.	Description	Unit	Qty.
7.8	150 mm dia (Tank outlet, Suction header, Interconnection etc.) PN-10 .	Nos.	2.00

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Item Nos.	Description	Unit	Qty.
8	Providing, fixing, testing and commissioning of PRV in water supply downtake / branch pipes. To be installed inside shaft, on-line type.		
	Additional supports shall be provided to PRV as per site condition.		
	Brass insert (male / female) adaptors shall be used for all PRV's. Also 'Tee' used before & after PRV for pressure gauge shall be brass insert type.		
8.1	15 mm dia	Nos.	0.00
8.2	20 mm dia	Nos.	0.00
8.3	25 mm dia	Nos.	92.00
8.4	32 mm dia	Nos.	68.00
8.5	40 mm dia	Nos.	0.00
8.6	50 mm dia	Nos.	0.00
9	Providing, fixing, testing and commissioning water meter with direct reading dial in KL with all integral parts of gun metal or brass with necessary fitting such as threaded pieces, unions pressure gauge and isolation cock, flanges piece for future removal, flanges / unions complete with all necessary testing charges and obtaining test certificates from municipal authorities, on following size pipe lines. (Municipal water , STP, Bore well & DCB lines). (ball valve & strainer considered in respective item)		
9.1	25 mm dia	Nos.	1.00
9.2	50 mm dia	Nos.	2.00

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Item Nos.	Description	Unit	Qty.
9.3	80mm dia. (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (STP, DCB line if required) .	Nos.	1.00
10	Providing & fixing Auto Air vent for cold water supply risers, suitable for pressure not less than 15 Kg/Sq.cm. (For top 3 floor loop) .		
10.1	15 mm dia	Nos.	8.00
10.2	25 mm dia	Nos.	0.00
11	Providing & fixing of heavy quality Non Return valves of gun metal / brass / CI body with screwed / flanged ends pressure as mentioned below & as per details mentioned in approved materials list etc. complete (For plant room, sump pump, etc.) Flanged end - Dual plate type .		
11.1	65 mm dia (Main Flushing pump) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-16) .	Nos.	3.00
11.2	100 mm dia (Main Domestic pump) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-16) .	Nos.	0.00
11.3	80 mm dia (DCB / sump pump / Tanker / STP) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-10) .	Nos.	2.00
11.4	80 mm dia (Main Domestic pump)) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-16) .	Nos.	3.00
11.5	100 mm dia (Irrigation / sump pump) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-10)	Nos.	2.00

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Item Nos.	Description	Unit	Qty.
12	Providing & fixing of heavy quality Strainers (with 400 micron SS mesh) of gun metal / brass / CI body with screwed / flanged ends 16kg class as per details mentioned in approved materials list etc. complete (For Terrace Loop, Water meter etc.) . CI body Strainers with anti corrosive coating.		
	Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers for flanged end strainers.		
12.1	25 mm dia. (for water meter) (GM / Brass - Y Type strainer)	Nos.	1.00
12.2	50 mm dia (for water meter) (GM / Brass - Y Type strainer)	Nos.	2.00
12.3	50 mm dia (for loop / pump suction) (VARIE - ECS B).	Nos.	0.00
12.4	65 mm dia (terrace loop) (VARIE - ECS B).	Nos.	2.00
12.5	80 mm dia (terrace loop) (VARIE - ECS B).	Nos.	2.00
12.6	65 mm dia (for water meter, pump suction - flange type).	Nos.	2.00
12.7	80 mm dia (for water meter, pump suction - flange type).	Nos.	3.00
12.8	100 mm dia (for terrace loop / pump suction / water meter - flange type)	Nos.	5.00
13	Providing and fixing 600 mm dia cast iron (heavy duty) water tank circular cover with frame 40 mm CI seat & lockable arrangement complete in all regards (Total WT of cover and frame to be not less than 20.8 kg). Including painting with 3 coats of bitumastic paint. (All UG & OH tanks).	Nos.	15.00

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Item Nos.	Description	Unit	Qty.
14	Providing & Constructing of block masonry chamber (for housing valves, Meters etc.) of block work with cement mortar 1:5 (1 cement:5 fine sand) plastering internal face with cement mortar 1:3 (1 cement:3 fine sand and rough plaster on outer face with a floating coat of neat cement with required duty Ductile Iron manhole cover and frame with locking arrangement, Including excavation, dewatering, refilling, watering, ramming and removing the surplus excavated material complete as required. (All sizes are clear internal sizes).		
14.1	Size 900 x 600 x 900 mm deep with 900 x 600 Heavy duty D.I. frame and cover. (104kg / C-250 / MR 17)	Nos.	2.00
14.2	Size 900 x 450 x 900 mm deep with 900 x 450 Medium duty D.I. frame and cover. (67kg / B-125 / MR 18)	Nos.	0.00

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Item Nos.	Description	Unit	Qty.
15	Providing & fixing of vapour vent pipes of 50mm with GI Tee, Elbow, Pvc Mosquito proof jali including required jointing material etc. complete. (1 set = 2" pipex300mm + 1 Coupling + 1 Tee + 2 Elbow + 2 mosquito jali) .		
15.1	50 mm dia	Set	24.00
16	Providing, fixing, jointing and testing in position approved quality of dial type pressure gauge with glycerin filled S.S. Bourdon tube pressure scale of 0 to 10 / 15 / 25 kg / cm ² . Application areas (activity) - A) Pump room delivery lines (flushing / domestic) B) Downtake PRV setting (4 gauges min.) C) For Terrace Booster system.		
	Calibration shall be done for pressure gauges and certificate shall be submitted before starting of pressure testing works for any plumbing lines and for pressure gauges for pump rooms. (as per requirement of ISO) With below details - a) Make of gauge - b) Pressure Range of gauge - c) Date of calibration - d) Calibration DUE on -	Nos.	12.00
17	Providing, fixing, jointing and testing in position double flanged flexicon rubber expansion joint for Domestic & Flushing pump delivery lines + between third NRV & Butterfly valve (suitable for system test pressure) of standard length as per manufacturers specs including rubber gaskets, GI flanges, GI nuts, bolts and washers complete as required. PN-16		
17.1	65mm dia.	Nos.	3.00
17.2	80mm dia.	Nos.	3.00
17.3	100mm dia. (PN-10, for irrigation pump)	Nos.	2.00

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Item Nos.	Description	Unit	Qty.
18	Providing & fixing of heavy quality Ball type Non Return Valve of CI body with flanged ends 10kg class, Nitrile rubber coated ball with inside lining of Rubber for valve. Including rubber gasket, G.I. flanges, GI nuts, GI bolts, GI washers. (For Flushing Tanker inlet / Sewage pump delivery).		
18.1	80mm dia.	Nos.	0.00
18.2	100mm dia.	Nos.	0.00
TOTAL : "B" CARRIED TO SUMMARY			

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
C.	INTERNAL DRAINAGE (SOIL, WASTE & VENT)		
1	Providing, fixing, jointing, testing and commissioning of UPVC pipe Type-A for R.W. upto 160mm dia. confirming to IS 13592 & AGRI for R.W. above 160 mm dia confirming to IS:4985 cut to required lengths including all necessary fittings and specials such as 45 or 90 deg. bends, Tee's, Y's, access pipe with solvent cement jointing. Fixing at wall / ceiling level supported by galvanized brackets & pvc coated clamps & hangers etc.		
	All pipe diversions shall be done with 45 degree bends where ever required due to beam offsets or cutout changes at any level. For vertical support, fix 4 brackets/floor or 1 m c/c whichever is less.		
	For Vertical Rain Water Pipes		
	Scope includes connection of service floor drains to Multiples RWP pipes with using Y' & Bends.		
1.1	160 dia (Type A) Pasted type.	RM	325.00
1.2	110 dia (Type A) Pasted type.	RM	350.00
1.3	75 dia (Type A) Pasted type (for small terraces of watchmen cabin or other structure etc. as per site condition).	RM	10.00
2	Providing, fixing, jointing, testing and commissioning of UPVC pipe Type-A for vent & Type-B for Soil, Waste & A.C. drains upto 160mm dia. confirming to IS 13592 cut to required lengths including all necessary fittings and specials such as as 45 or 90 deg. bends, Swept Tee's , Y's, access pipe with rubber ring jointing. PVC Cowl on top with solvent joint.		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
	Fixing at wall / ceiling level supported by galvanized brackets & PVC coated clamps & hangers etc. Vertical stacks shall be truly vertical & parallel to the wall and supported on M.S. Hot dip Galvanised brackets with PVC coated 'U' clamps, so that the pipes are at least 100mm away from the finished surface/ plaster. Also 75mm to 100mm gap between 2 pipes should be maintained. All pipe diversions shall be done with 45 degree bends where ever required due to beam offsets or cutout changes at any level.		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
	For vertical support, fix 3 brackets/floor or 1 m c/c whichever is less.		
	For Vertical Soil, Waste, Vent, Anti Siphonage, AC Drain etc.		
	Only Swept Tee with Door shall be used for all Soil and Waste lines which are with Tee connections shown on drawing.		
2.1	160 dia (Type B)	RM	0.00
2.2	110 dia (Type B) (Soil pipe).	RM	1284.00
2.3	75 dia (Type A) (Toilet - Anti Siphonage main & branch pipe / Vent for starting point).	RM	1545.00
2.4	75 dia (Type B) (Toilet / Kitchen Waste pipe).	RM	1926.00
2.5	75 dia (Type B) (Balcony drain pipe).	RM	642.00
2.6	75 dia (Type A) (AC Drain pipe). (With provision of single Y + 45 deg. Bend on every floor)	RM	642.00
3	Providing and fixing including labour for fixing in position PVC Floor trap with 75mm dia outlet and a trap seal of 50mm for kitchen and bath wet area & 20 to 25mm for dry area. If two or three floor trap connections are with 'Y' or separately connected with waste lines, then all floor traps shall be of 50mm water seal. Making proper connection, cutting chase / hole in floors /slabs and bringing the same in proper condition and shape after placing the trap in right position complete as required with encasing in CC M-15.		
3.1	100 mm inlet and 75 mm outlet. (UPVC) 50mm water seal for toilet dry / wet area of 250mm sunk toilets.	Nos.	17.00
3.2	100 mm inlet and 75 mm outlet. (UPVC) 50mm water seal for toilet wet area of 50mm sunk toilets (suspended with bottom support).	Nos.	382.00
3.3	100 mm inlet and 75 mm outlet. (UPVC) 50mm water seal for kitchen or pantry area.	Nos.	199.00

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Item Nos.	Description	Unit	Qty.
3.4	100 mm inlet and 75 mm outlet. (UPVC) Regular water seal for balcony drain.	Nos.	199.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
4	Providing, fixing, jointing, testing and commissioning of UPVC pipe Type-B for Soil, Waste upto 160mm dia. confirming to IS 13592 cut to required lengths including all necessary fittings and specials such as as 45 or 90 deg. bends, Tee's, Y's, access pipe with solvent cement jointing. (For Pipes buried in flooring).		
	Setting the pipes in cement concrete M-15, 100mm all around of pipes as per specifications after testing of pipes. SWR range pasted pipes (Type-B) & fittings (without rubber rings) with solvent joints for all sunken area (concealed) piping for toilets / kitchens (Water testing of pipes to be done after 12 hours, after last solvent cement joint. Where ever possible use Aqua Drain test plugs for testing of lines.) Including civil material for pcc+encasing (concrete quantity considered separately).		
4.1	110 dia (Type B) (Toilets / Balcony).	RM	90.00
4.2	110 dia (Type B) (Kit.).	RM	70.00
4.3	75 dia (Type B) (Toilets)	RM	10.00
4.4	75 dia (Type B) (Kit. & Balcony)	RM	200.00
5	Providing, fixing, jointing, testing and commissioning of UPVC pipe Agri. range 6 kgf pipes and fittings for 40, 50 & 63mm dia. cut to required lengths including all necessary fittings and specials such as as 45 or 90 deg. bends, Tee's, Y's, access pipe with solvent cement jointing. Vertical pipes (above sunk level) in necessary wall chases and finishing of same with geru mixed cement mortar. Including civil material for pcc+encasing (concrete quantity considered separately).		
5.1	Setting the pipes in cement concrete M-15, 100mm all around of pipes as per specifications after testing of pipes. (Water testing of pipes to be done after 12 hours, after last solvent cement joint. Where ever possible use Aqua Drain test plugs for testing of lines.) .		
	For Pipes Buried in Toilets / kit. from Wash Basin, Sink, etc to Floor trap / Main Drain (50mmx25mm reducer socket and 40mmx25mm reducer socket shall be fixed for washing machine / dish washer outlet).		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
5.1.1	40 mm dia (Toilets)	RM	40.00
5.1.2	40 mm dia (Kitchens)	RM	400.00
5.1.3	50 mm dia (Toilets)	RM	0.00
5.1.4	50 mm dia (Kitchens)	RM	10.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
5.2	Providing and fixing including labour for fixing UPVC pasted pipes (Type-B) & fittings (without rubber rings) with solvent joints of 75 & 110mm dia. and Agri. range 6 kgf pipes and fittings for 40, 50 & 63mm dia. for all suspended horizontal piping for toilets / kitchens including all necessary fittings and specials such as as 45 or 90 deg. bends, Tee's, Y's, access pipe etc.		
	Pipe portion going in concrete area shall be applied with solvent and fine sand. (Water testing of pipes to be done after 12 hours, after last solvent cement joint. Where ever possible use Aqua Drain test plugs for testing of lines.)		
	Supports shall be Split clamp with rubber lining for entire suspended pipings as per specs. Split clamp rod & Anchor bolt dia. – 8mm for 40mm & 50mm pipes and 10mm for 63mm, 75mm and 110mm pipes. Additional supports to be provided near fittings if required as per site condition & instructions of engineer in-charge. Horizontal supports spacing (Center to Center) shall be as follows - 40mm = 400mm / 50mm= 500mm / 75mm= 750mm / 110mm = 1000mm / 160mm & above = 1500mm.		
	For suspended pipes in Toilets / kitchens from Wash Basin, Sink, Washing Machine, Dish Washer etc to 'P' Trap. Supports required at bottom of 'P' Trap & Floor Trap.		
	Vertical pipes (above slab level) in necessary wall chases and finishing of same with geru mixed cement mortar.		
	Horizontal SWR drains for 50 / 65mm sunk toilets and Kitchens (suspended piping) (50mm x 25mm reducer socket and 40mm x 25mm reducer socket shall be fixed for washing machine / dish washer outlet).		
5.2.1	40 mm dia	RM	650.00
5.2.2	50 mm dia	RM	0.00
5.2.3	63 mm dia	RM	0.00
5.2.4	75 mm dia	RM	185.00
5.2.5	110 mm dia	RM	660.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
6	Providing, fixing, Testing and Commissioning uPVC agricultural pipe with necessary fittings, GI brackets, clamps, split clamp with rubber etc. complete. Concealed / false ceiling area pipes if any, with elastomeric rubber insulation (clamps shall be fixed after insulation) and exposed area pipes without insulation & supports at 1 mtr. c/c. (For A.C. Drain connection in shaft, 'U' trap to be done with using fittings before connecting to vertical pipe) Quantity to be check & order as per site condition.		
6.1	25 mm dia (8kgf or 10kgf). (For 250mm sunk area sleeve)	RM	170.00
6.2	20 mm dia (8kgf or 10kgf)	RM	0.00
7	Providing, fixing, jointing, testing and commissioning of heavy quality approved make CI spigot and socket pipes for soil, waste and vent, conforming to IS 3989 including all fittings like offsets, 45 deg. bends, single or double junctions of different degrees, with access doors, etc. including providing & fixing with caulking spun yarn and DRIP seal jointing with testing of pipes and making the joints leak proof. including fixing them vertically against walls on M.S. Hot dip Galvanised brackets & G.I. 'U' clamps or hangers including all scaffolding, testing after laying the pipes as specified etc. Complete.		
	Vertical SWR drains in shafts for toilets / kit. (ground) with C.I. Cleaning pc. at bottom. (where applicable).		
7.1	80 mm dia	RM	0.00
7.2	100 mm dia (including canopy drain).	RM	8.00
8	Providing and fixing P.V.C. WC connector with rubber lip ring to suit WC outlet and proper connection complete as required .		
8.1	For Servants / drivers toilets (Regular)	Nos.	1.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
9	Providing and fixing including labour for fixing in position PVC 'P' (110mm x 75mm) Trap with necessary support at bottom. Connection with pvc puddle sleeve provided with solvent joint etc. complete.		
	For toilet shaft drains	Nos.	1.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
10	Providing, fixing, Testing and Commissioning of uPVC Puddle Flanges (Type-B pipe only) including fixing with BCP screws (35 x 8), 4 nos. per flange. Center of flange shall be as per w.c. outlet & 'P' Trap outlet dimensions and same shall be in plumb before fixing screws.		
10.1	110mm dia x 600mm long (For w.c. outlet, floor drain & 'P' Trap etc.)	Nos.	764.00
10.2	50mm dia x 600mm long (For dish washer / W.M. outlet etc.)	Nos.	0.00
10.3	40mm dia x 800mm long (For basin outlet / sink outlet / W.M. outlet etc.).	Nos.	382.00
11	Providing, Fixing, Testing & Commissioning of C.I. rain water Grating.		
11.1	100mm dia. Round (dome shape) for 75 mm outlet.	Nos.	0.00
11.2	150mm dia. Round (dome shape) for 110 mm outlet.	Nos.	4.00
11.3	225mm dia. Round (dome shape) for 160 mm outlet .	Nos.	0.00
11.4	200 x 200 square for 160 mm outlet.	Nos.	8.00
11.5	150 x 150 square for 110 mm outlet.	Nos.	8.00
11.6	300 x 300 square for 250 mm outlet.	Nos.	0.00
TOTAL : "C" CARRIED TO SUMMARY			

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
D.	EXTERNAL SEWAGE / RAIN WATER DRAINAGE		
1	Excavation in trenches for pipelines including trimming bottom and sides, cutting for sockets, ramming of bottom and disposal of surplus earth etc.		
1.1	Depth of Excavation not exceeding 1.5m (Excavation in all classes of soil).	Cum	250.00
1.2	Extra over 1.1 above for depths exceeding 1.5m but not greater than 2.5m (Excavation in all classes of soil).	Cum	110.00
1.3	Excavation in slush.	Cum	0.00
1.4	Excavation in hard rock / existing concrete. (refilling in 200mm layers with selected excavated soil)	Cum	0.00
1.5	Providing and laying cement concrete M-15 as per specifications for PCC / Encasing / Foundation etc. (including 250mm sunk toilet / kitchen / balcony pipe encasing)	Cum	80.00
1.6	Backfilling of trenches with murum / soil.	Cum	200.00
2	Providing, fixing, jointing and testing in position the following UPVC pipes confirming to IS: 4985 of 6 Kg / Sq.cm for drainage system cut to required length. Pipe to be laid below ground level in trenches upto required depth, dewatering, jointing of pipes with solvent cement, water testing of joints after 1 hour and setting the pipes in cement concrete M-15 as per specifications 150mm thk. all around, refilling of trenches, and making good the same complete as required.		
	Scope of work includes piping and manholes upto existing manhole or municipal manhole on road and connection with breaking the wall of municipal / road manhole and making good the same with Cement mortar 1:3, finished with a floating coat of neat cement and making the same good etc. complete.		
	For RWP System from Plumbing duct to RW chamber, Drain chamber Branch pipes.		

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Item Nos.	Description	Unit	Qty.
	Including civil material for pcc+encasing (concrete quantity considered separately)		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
2.1	200 mm dia	RM	0.00
2.2	160 mm dia	RM	25.00
2.3	110 mm dia	RM	60.00
2.4	75 mm dia	RM	20.00
3	Providing, fixing, jointing and testing in position the following Double wall corrugated H.D.P.E. pipes, for non pressure storm water and sewerage applications. (As per ISO 21138 – 3) for drainage system cut to required length.		
	Pipe to be laid below ground level in trenches upto required depth, dewatering, jointing of pipes with solvent cement, water testing of joints after 1 hour and setting the pipes in cement concrete M-15 as per specifications 150mm thk. all around, refilling of trenches, and making good the same complete as required.		
	Scope of work includes piping and manholes upto existing manhole or municipal manhole on road and connection with breaking the wall of municipal / road manhole and making good the same with Cement mortar 1:3, finished with a floating coat of neat cement and making the same good etc. complete.		
	For RWP System from building to SWD, RWH Piping, Building Drainage etc. - Main and Branch pipes.		
	DWC SN4 pipes shall be used up to 1.5mtr. deep drain lines and SN8 pipes shall be used if depth is more than 1.5 mtr.		
	Including civil material for pcc+encasing (concrete quantity considered separately)		
3.1	100 mm Int. dia (SN-4)	RM	0.00
3.2	150 mm Int. dia (SN-8)	RM	80.00
3.3	200 mm Int. dia (SN-8)	RM	80.00
3.4	250 mm Int. dia (SN-8)	RM	20.00
3.5	300 mm Int. dia (SN-8)	RM	25.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
4	Providing and fixing round mouthed pvc Bottle gully trap with 300mm x 300mm chamber with CI cover at top including embedding the trap in cement concrete M-15 construction of brick masonry chamber internal clear size 300 x 300 mm in brick masonry 250mm thick in cement sand mortar 1:4 over M-15 cement concrete bed 150mm thick, plastering inside outside and all exposed surfaces with 1:3 cement sand plaster 15mm thick.		
	Including providing and fixing of CI frame & cover size 300 x 300mm weighing together not less than 7.8Kgms painting all CI items with 3 coats of approved bituminastic paint etc. complete.		
	For all waste lines including AC / Balcony / Shaft drain lines etc.	Nos.	21.00
5	Providing & Constructing block masonry inspection chamber of below sizes and depth upto 1375mm including the following: a) 200 thick block masonry walls in CM 1:4 b) C.C. foundation bed in M-15, 150 mm thick size 1600 x 1150 c) Cement plaster inside and outside in CM 1:3 (with cement ghotai from inside)		
	d) CC M-15 in haunches and channels finished smooth in CM 1:2 e) C C M-15 cap on top 200 thick f) Providing & fixing of Ductile / Cast iron cover & frame g) 75 wide fillet (wata) all round block / brick masonry and c.c. bed in CM 1:3 h) All excavation in any type of soil removal and disposing off surplus earth after backfilling as directed including dewatering.		
	All civil materials in contractors scope.		
5.1	900mm x 450mm		
5.1.1	with 80 kg D.I. Cover (Item code no = C-250 / MR 14)	Nos.	9.00
5.1.2	with 67 kg D.I. Cover (Item code no = B-125 / MR 18)	Nos.	0.00
5.1.3	with 50 kg D.I. Cover (Item code no = A-15 / MR 4)	Nos.	0.00
5.2	900mm x 600mm		
5.2.1	with 104 kg D.I. Cover (Item code no = C-250 / MR 17)	Nos.	2.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
5.2.2	with 84 kg D.I. Cover (Item code no = B-125 / MR 19)	Nos.	0.00
5.2.3	with 65 kg D.I. Cover (Item code no = A-15 / MR 7)	Nos.	0.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
5.3	600mm x 600mm		
5.3.1	with 62 kg D.I. Cover (Item code no = C-250 / MS 24)	Nos.	8.00
5.3.2	with 58 kg D.I. Cover (Item code no = B-125 / MS 23)	Nos.	0.00
5.3.3	with 40 kg D.I. Cover (Item code no = A-15 / MS 6)	Nos.	0.00
6	Providing & Constructing brick masonry conical manholes including the following: a) Bed PCC for chamber to be M-15 concrete 200 mm thick, with 6" projection. b) Chamber to be constructed using bricks 225mm thick wall. c) Providing & fixing of Ductile / Cast iron cover & frame d) Plastic steps to be provided. (Copolymer Polypropylene, Steel reinforced manhole step, is injection molded around a 12mm dia. IS-1786 grade Fe-415 steel reinforcing bar.)		
	e) Size of chamber - 1200 mm (for depths > 1375mm and < 2285mm), or 1500 mm (for depth > 2285mm) at bottom, and 0.53 m. at top plaster 1:1 (out) and 1:1 (inside) block masonry 1:3 f) Haunching & benching M-15. g) Depth beyond 2285mm should be in double block 200mm cylindrical. h) C C M-15 cap on top 300 thick i) Cement plaster inside and outside in CM 1:3 (with cement ghotai on both sides)		
	All civil materials in contractors scope.		
6.1	Depth upto 2285mm (1200mm dia. Clear at bottom)		
6.1.1	with 60 kg D.I. Cover (Item code no = C-250 / MC 25) 600mm dia.	Nos.	8.00
6.1.2	with 53 kg D.I. Cover (Item code no = B-125 / MC 24) 600mm dia.	Nos.	0.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
6.1.3	with 42 kg D.I. Cover (Item code no = A-15 / MC 12) 600mm dia.	Nos.	0.00
6.2	Depth beyond 2285mm (1500mm dia. Clear at bottom) in double block 200mm cylindrical.		
6.2.1	with 60 kg D.I. Cover (Item code no = C-250 / MC 25) 600mm dia.	Nos.	0.00
6.2.2	with 53 kg D.I. Cover (Item code no = B-125 / MC 24) 600mm dia.	Nos.	0.00
6.2.3	with 42 kg D.I. Cover (Item code no = A-15 / MC 12) 600mm dia.	Nos.	0.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
7	Providing, Fixing, testing & commissioning of Nu-Drain UG Ultra Inspection chambers of Supreme including the following: a) Bed PCC for chamber to be M-15 concrete 100 mm thick, with 6" projection. b) Encasing of Inspection Chamber / riser pipes (100mm all around) with CC M-15. c) Providing & fixing of Necessary chamber cover & frame		
	All civil materials in contractors scope.		
7.1	Left hand 45 deg. and 90 deg. Junction (315mm x 110mm x 110mm x 110mm) including below items.	Nos.	2.00
	a) 315mm dia. x 154mm ht. Riser with rubber seal (2 or 3 nos. as per depth)		
	b) 315mm x 315mm Frame & Cover (Heavy duty)		
7.2	Right hand 45 deg. and 90 deg. Junction (315mm x 110mm x 110mm x 110mm) including below items.	Nos.	3.00
	a) 315mm dia. x 154mm ht. Riser with rubber seal (2 or 3 nos. as per depth)		
	b) 315mm x 315mm Frame & Cover (Heavy duty)		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
8	Providing, Fixing, Testing, & commissioning of Hubless centrifugally cast iron pipes and fittings (Epoxy coated inside & outside) as per ISO-6594 / IS-15905 Standard and jointing with ASTM C1277 marked stainless steel shielded coupling with a double stainless steel bolt & Screw Housing Incorporating a elastometric rubber gasket , system having noise level certification of not more than 25db(A) at 2L/S & inclusive of all necessary specials like bends, tees, offsets, junctions, cowl end plug, inspection pipe, etc. laid under floor/fixed on walls and in pipe shafts. The rate shall be including angle supports, hanger, nuts bolts, anchor fasteners, fixing clamps/ channels with U-bolts, (spacing and details mentioned below) etc. complete. Cutting, chases/holes in floors / walls / slab. (For Soil & Waste Water Pipes upto 300mm Dia. Exposed in Basements)		
	Scope includes, connections of CI pipes with G.I. Puddle flanges provided in retaining wall, including P & F of necessary fittings like thimble pc., tail pc., G.I. flanges, rubber packing, G.I. / S.S. nut-bolt-washer, dripseal joints etc. complete with all respect.		
	Soil line branches shall be connected to main line with using two no. 45 deg. bends. Direct connections with 90 deg. bends are not allowed. (i.e. For all Vertical Soil line connections with Suspended drain lines)		
	Horizontal support spacing for 80mm = 1.0mtr / 100mm = 1.2mtr / 150mm = 1.4mtr / 200mm and higher dia. 1.5mtr. center to center for straight lines.		
	Hanger rod dia. for support clamps – 80 & 100mm = 12mm / 150 & 200mm = 16mm / 300mm = 19mm.		
	(For Suspended drain lines - main & branch at stilt & pump room ceiling piping)		
	Fix Fabricated G.I. Bracket with G.I. 'U' clamps near plumbing shaft (i.e. immediate diversion from vertical line to horizontal line), near bends of main header, near branch tapping with header. (Use 50x50x5 Angle for brackets) and every 5 mtr. Interval for horizontal piping.		
8.1	80mm dia.	RM	3.00
8.2	100mm dia. (canopy drain if required).	RM	60.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
8.3	150mm dia.	RM	90.00
8.4	200mm dia.	RM	0.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
9	Providing, fixing, Testing and Commissioning of CI spun 'P' traps for waste line connections with GI brackets and DRIP seal jointing with testing of joints for leak proof. Waste line branches shall be connected to main line with using single 'Y' & 45 deg. bends. Direct connections with 90 deg. bends are not allowed for waste lines.		
	Vertical Waste line connections with Suspended drain lines.		
9.1	80 mm inlet and 80 mm outlet. (for AC drain)	No.	2.00
9.2	100 mm inlet and 100 mm outlet. (Toilet / Kit. / Duct / Balcony Waste line)	No.	13.00
10	Providing, fixing, Testing and Commissioning of UPVC / PP manually operated backflow preventer (NRV) for rain water lines discharging to main SWD / Nalla from DCB. Including construction of valve chamber of 900 x 600 size as per company specifications and P & F of Heavy duty D.I. cover 104 kg (Item code no = C-250 / MR 17), including excavation in any type of soil . (Chamber / civil items considered separately in respective items)		
	(To be confirm before order)		
10.1	150mm dia.	Nos.	1.00
10.2	200mm dia.	Nos.	0.00
TOTAL : "D" CARRIED TO SUMMARY			

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
E.	PUMPS AND EQUIPMENTS		
1	Supply, Installing, testing and commissioning of Water Transfer Pump for OH Tank Filling Vertical inline multistage centrifugal pump. Each pump shall have TEFC 2900 RPM, IE3 Rating efficiency, three phase electric motor. The pump shall have cast Iron head & base with SS Shaft/ Impeller/ Intermediate chamber. Pump shall be provided with mechanical seal. The pump shall be suitable for auto/ manual operation.		
	Accessories like NRVs, isolating valves, suction and delivery header / piping, pressure gauge, MS base frame (Synthetic enamel painted over a coat of primer) has to be provided. (Quantities of Valves / pipe / gauges etc. considered in respective BOQ items).		
	Making Suitable cement concrete foundation with cement plaster for pumps with supply & fixing of required Anti vibration rubber pads (150mm x 150mm x 25mm thk. = 4 nos. per pump) and foundation bolts for pumps Bolt Sizes : a) Upto 10 HP Pump - 12mm Ø, b) 10 HP to 50 HP Pump - 16 mm Ø, c) Above 50 HP Pump - 20 mm Ø, d) As per pump manufacturers requirement.		
1.1	Flow rate : 535.33 LPM		
	Head : 103 Mts		
	Purpose : Domestic Water transfer (1 Set = 1W+1SB)	Set	1.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
	Pump selected are, Kirloskar make Multistage Vertical inline pump , Model – KCIL32-08-2, 8 stages, 77.73% efficiency, 2900 RPM, Pump coupled with 20 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 32120 LPH / 535.33 LPM / 8.92 LPS. at 103 meter head. OR Lubi make Multistage Vertical inline pump , Model – LCR32-08-2, 8 stages, 75.93% efficiency 2900 RPM, Pump coupled with 20 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 32120 LPH / 535.33 LPM / 8.92 LPS. at 103 meter head. OR Wilo Mather + Platt make Multistage Vertical inline pump , Model – HELIX FIRST V3606/2-5/25/E/KS/415-50, 2900 RPM, Pump coupled with 20 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 32120 LPH / 535.33 LPM / 8.92 LPS. at 103 meter head.		
1.2	Flow rate : 268 LPM		
	Head : 99 Mts		
	Purpose : Flushing Water transfer (1 Set = 1W+1SB)	Set	1.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
	Pump selected are, Kirloskar make Multistage Vertical inline pump , Model – KCIL 15-09, 9 stages, 70.99% efficiency, 2900 RPM, Pump coupled with 10 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 16080 LPH / 268 LPM / 4.46 LPS. at 99 meter head. OR Lubi make Multistage Vertical inline pump , Model – LCR15-09, 9 stages, 71.46% efficiency 2900 RPM, Pump coupled with 10 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 16080 LPH / 268 LPM / 4.46 LPS. at 99 meter head. OR Wilo Mather + Platt make Multistage Vertical inline pump , Model – HELIX FIRST V1610-5/25/E/KS/415-50, 2900 RPM, Pump coupled with 10 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 16080 LPH / 268 LPM / 4.46 LPS. at 99 meter head.		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
2	Supply, Installing, testing and commissioning of Skid mounted terrace booster system with suction & delivery header complete including necessary valves, NRV's, air vessel, electrical panel, dry run protection etc. Each pump shall have TEFC 2900 RPM, IE3 Rating efficiency, three phase electric motor. The pump shall have cast Iron head & base with SS Shaft/ Impeller/ Intermediate chamber. Pump shall be provided with mechanical seal. System shall operate on Auto mode. (other than skid mounted system Valves / pipe / gauges etc. quantities are considered in respective BOQ items).		
	Making Suitable cement concrete foundation with cement plaster for pumps with supply & fixing of required Anti vibration rubber pads (150mm x 150mm x 25mm thk. = 4 nos. per pump) and foundation bolts for pumps Bolt Sizes : a) Upto 10 HP Pump - 12mm Ø, b) 10 HP to 50 HP Pump - 16 mm Ø, c) Above 50 HP Pump - 20 mm Ø, d) As per pump manufacturers requirement.		
	To be order only after confirmation from HOH team.		
2.1	Flow rate : 215 LPM		
	Head : 15 Mts		
	Purpose : Terrace booster System for DOMESTIC (1 Set = 1W+1SB)	Set	0.00
	Pump technical details to approve from HOH MEP before order.		
3	Fixing, jointing and testing including labour for fixing in position Irrigation pumps with supply and fixing of required Anti vibration rubber pads (150mm x 150mm x 25mm thk. = 4 nos. per pump) and foundation bolts for pumps Bolt Sizes : a) Upto 10 HP Pump - 12mm Ø, b) 10 HP to 50 HP Pump - 16 mm Ø, c) Above 50 HP Pump - 20 mm Ø including making concrete foundation for pumps. Valves and piping considered in respective BOQ items.		
3.1	Irrigation system pumps (1W+1SB)	Set	1.00
	TOTAL : "E" CARRIED TO SUMMARY		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
F.	ELECTRICAL PANELS AND CABLES		
1	Design, fabrication, assembling, wiring, supply, installation, testing and commissioning of following Wall mounted / Floor mounted (including base stand 200 mm height) motor control centre, shall be fabricated out of 14 gauge CRCA sheet steel in form in 3b formation with reinforcement of suitable size angle iron, channel 'T' sections irons and/or flats wherever necessary. Cable gland plates shall be provided on top as well as at the bottom of the panels. The indoor panels shall be IP 42 and outdoor panels shall be IP 54 type. The Panel shall have proper space with required clearance for cables, incoming / outgoing switchgear, CTs, PTs, ammeter, Voltmeter, selector switches, Insulated and properly supported compartmentalized bus bars with heat shrinkable sleeves along with the other control circuit accessories within the panel and any other electrical component mentioned in the Panel SLD.		
	The Panel shall have seven tank pre treatment process comprising of degreasing, rinsing, de-rusting, rinsing, phosphatising, rinsing and passivation followed by powder coat painting having a paint thickness of 60 microns or as specified. The Panel shall be Dust/Vermin proof. Painting as per specifications with 2 coats of red oxide primer and final approved shade of powder coated paint. 2 Nos. earthing terminals shall be provided for 3 phase, 4 wire, 50 Hz supply system. Lifting hooks shall also be provided in case of large panels. as per attached SLD & BOM.		
	Water Level sensor Shall be there in UG and overhead tanks to get Pumps on off Command to the system.		
1.1	MCC –PLUMBING PANEL (plant room)		
	Incomer		
	160A TP+N MCCB with MP Based OL,SC,ST Releases, Aux. 25KA & RHOM		
	R Y B, Indicating Lamps		
	Off On Trip Indicating Lamp		
	Digital Multi Data Meter with RS 485 Port BMS Compatible		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
	Current Transformer 150/5A CL.-1 5VA		
	Digital Earth Leakage Relay with CBCT		
	Control MCB 6A TP 10KA C Curve		
	Necessary aux. contact of the I/C MCCB's Off,On,Trip status to BMS		
	Bus Bar		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
	160 amps TPN (25KA) aluminium bus bar with heat shrinkable insulation sleeves.		
	Outgoings		
a.	2 Nos. TP MPCB with 20 HP Star delta starter as per type -2 cordination (1 set - 1W+1SB) as outgoing feeder to Domestic transfer pump . Each compartment shall contain CT operated digital electronic ammeter of 0-50 amps range with auto/manual selector switch, on/off push button ,SPP and indicating lamps with MCB's , 'ON/OFF/TRIP' status of motor.		
b.	2 Nos. TP MPCB with 10 HP Star delta starter as per type -2 cordination (1 set - 1W+1SB) as outgoing feeder to Flushing transfer pump . Each compartment shall contain CT operated digital electronic ammeter of 0-50 amps range with auto/manual selector switch, on/off push button ,SPP and indicating lamps with MCB's , 'ON/OFF/TRIP' status of motor.		
c.	(Provision) 2 Nos. TP MPCB with 2.0 HP Direct on line starter as per type -2 cordination (1 set - 1W+1S) as outgoing feeder to Skid mounted terrace Booster System pumps for Domestic . Each compartment shall contain CT operated digital electronic ammeter of 0-10 amps range with auto/manual selector switch, SPP and indicating lamps with MCB's and 'ON/OFF/TRIP' status of motor		
	Spare MCCB's of following capacities:		
d.	63 amps TPN MCCB 1 No with TM based OL ,SC Release, Aux 25KA & RHOM with on off indicating lamp with control MCB, 6A SP 10KA C curve.		
e.	40 amps TPN MCB 1 No 10KA C Curve. - 01 nos		
f.	25 amps TPN MCB 1 No 10KA C Curve. - 01 nos		
g.	Necessary cable alleys, internal wiring, and interlocking, earthing for all equipment shall also included.		
	Notes:		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
a.	All MCCBs shall be of 25KA breaking capacity and suitable for motor duty application.		
b.	Provision shall be made for providing potential free contacts to all pumps starters for connection to building automation system.		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
c.	Provision shall be made for cyclic operation of Domestic & Flushing Pumps in auto mode. i.e. Standby pump shall also operate at an regular interval in auto mode, Sequence timer for pump operation shall be provided.		
d.	Provision shall be made to ensure system protection from dry running of the pump		
e.	All motor MCCB/protecting devices shall be as per TYPE -2 co-ordination.		
f.	Water level controller with sensor,float switch to be provided with Panel.		
	All Panels should be dispatched at site only after inspection at manufacturer workshop by representatives of HOH electrical department. Cost of traveling (i.e.- contractor to make to & fro arrangements for transportation), testing etc. shall be borne by the Contractor. Any Panel dispatched without inspection is bound for rejection.		
	MCC –PLUMBING PANEL AS DESCRIBED ABOVE	Set	1.00
3	Supplying & laying of following 1100 volt grade XLPE insulated aluminium / copper conductor armoured cables having outer sheath of fire retardent extruded (FRLS) compound as per specification in existing trenches, cable trays, ducts over bed of sand, clamped to wall with suitable clamps, saddles fixing bolts including connecting testing and commissioning. All cables to be labelled (black anodized) at both the ends.		
3.1	3c x 10 sq. mm copper armoured cable .	RM	100.00
3.2	3c x 4 sq. mm copper armoured cable .	RM	150.00
3.3	3c x 6 sq. mm copper armoured cable .	RM	100.00
3.4	3c x 2.5 sq. mm copper armoured cable .	RM	0.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
3.5	4c x 2.5 sqmm copper armoured cable.	RM	0.00
3.6	4c x 4 sqmm copper armoured cable.	RM	200.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
4	Supply & fixing of Cable end termination of the following XLPE insulated aluminium/copper conductor armoured cables of 1100 volt grade including supplying and fixing of bimettalic crimping lugs, double compression glands with earthing facility and cable sockets etc. complete as required.		
4.1	3c x 10 sq. mm copper armoured cable .	Nos.	8.00
4.2	3c x 4 sq. mm copper armoured cable .	Nos.	4.00
4.3	3c x 6 sq. mm copper armoured cable .	Nos.	8.00
4.4	3c x 2.5 sq. mm copper armoured cable .	Nos.	0.00
4.5	4c x 2.5 sqmm copper armoured cable.	Nos.	0.00
4.6	4c x 4sqmm copper armoured cable.	Nos.	2.00
5	Supply and fixing of following Earthing , bare GI/copper tapes / wires including all necessary fixing accessories and effecting connections as per specifications.		
5.1	25 x 3 mm. GI. Strip.	RM	160.00
5.2	10 Gauge Bare GI. Wire.	RM	100.00
5.3	08 Gauge Bare GI. Wire.	RM	160.00
6	Design, Fabrication, Supply, Unloading, Handling, Installation, Commissioning of Perforated ladder type Cable Trays of Pre galvanised 02 mm thick : As per IS: 2062 /1079 & Galvanising : As per IS: 2629/4759 ,all accessories include. Ladder type cable trays of following sizes, suspended from ceiling or fixed vertically on wall / column as per route directed at site, as approved complete as per technical specifications, drawings & relevant standards & site conditions.		
6.1	size : 150mm wide.	RM	60.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
6.2	size : 300mm wide.	RM	30.00
6.3	size : 450mm wide.	RM	0.00
6.4	100 x 50 mm GI trunking	RM	20.00
TOTAL : "F" CARRIED TO SUMMARY			

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
G.	MISCELLANEOUS ITEMS		
1	Providing and fixing G.I. blank flanges, making flanged joints with G.I. bolts, G.I. nuts, G.I. washers and rubber insertion 3 mm thick, testing the flange and joints etc. complete for pipes of following diameter. (for unused / spare plumbing sleeves of UG Tanks & Retaining wall)		
1.1	80 mm dia.	Nos.	1.00
1.2	100 mm dia.	Nos.	1.00
1.3	150 mm dia.	Nos.	1.00
2	Fabrication, Supply, installation & fixing of puddle flanges, 600mm long for the storage tanks. The puddle flanges shall be fabricated out of MS pipes and plates as per the drawing enclosed and hot dipped galvanized before installations for O.H. Tanks including Fire Fighting Tanks.		
2.1	50 mm dia (Screwed ends).	Nos.	17.00
2.2	80 mm dia (Flanged ends).	Nos.	2.00
2.3	100 mm dia (Flanged ends).	Nos.	8.00
2.4	150 mm dia (Flanged ends).	Nos.	0.00
3	Scaffolding for plumbing works in shaft (if required) with all safety measures for labourers working on scaffolding.	Sqft.	200.00

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
4	Pipe label works for all plumbing pipes (All Water supply lines - Flushing / Domestic / Solar etc. All Drainage lines - Soil / Waste / Vent / Rain water etc. All types of pump delivery lines and pumps, All UG & OH Tanks etc.) Labels should be positioned on the pipes so they can be easily read. Proper label placement is on the lower side of the pipe if the employee has to look up to the pipe, on the upper side of the pipe if the employee has to look down towards the pipe, or directly facing the employee if on the same level as the pipe. Labels should be located near valves, branches, where a change in direction occurs, on entry/re-entry points through walls or floors, and on straight segments with spacing between labels that allows for easy identification and as per requirement of engineer incharge.		
	For Label work with arrow marks – “Flushing tank”, “Domestic tank”, “Raw Water tank” & “Fire tank” etc. shall be marked on each cover respectively, after enamel paint, including label works for various areas as per pictures given in specifications, terrace - podium main loop & branch, down takes - uptake main & branch, pump delivery lines, pumps, sump pump lines, tank overflow / cleanouts, suspended CI drain lines, Suspended pvc drain or rain water pipes etc.		
4.1	All plumbing pipe label works as per above details as well as details and pictures shown in specifications. (TA+NTA).	Job	1.00
	All Toilets - Flushing / Domestic branch and All Kitchens - Domestic branch = 25% .		
	Terrace loop & branch pipes, All OH Tank capacity details, names on cover, tank overflow / cleanouts etc. = 15% .		
	Podium loop & branch pipes (where applicable), All UG Tank capacity details, names on cover, tank overflow / cleanouts etc. = 15%		
	Domestic, Flushing, Sump pump, other pump details, all pump delivery lines = 15%		
	Suspended CI Main & Branch drain lines = 15%		
	Suspended PVC Main & Branch drain lines (terrace / surface drain / garden drain) = 15%		
5	Supply and handover of Manual (Hand) Drain Cleaning Tool with steel coil.		

OAKWOOD BUILDING AT KALYAN

Item Nos.	Description	Unit	Qty.
	Material to be handover to HOH Plumbing department with necessary test and warrantee / guarantee ertificates of manufacturer.		
5.1	with 12 feet coil (Varie).	Nos.	2.00
5.2	with 25 feet coil (Varie).	Nos.	2.00
TOTAL : "G" CARRIED TO SUMMARY			

OAKWOOD BUILDING AT KALYAN

OAKWOOD BUILDING -Kalyan

SCHEDULE OF QUANTITIES

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
A.	SANITARY FIXTURES & FITTINGS (INSTALLATION)				
	Note:				
	i. Detail of Sanitary fixture are for the information of the Contractor, however model / makes of all sanitary fixture shall be selected by Architect / Interior designer / client and the same shall be binding for execution .				
	ii. All sanitary fixture and accessories / CP fittings shall be issued free of cost by the employer alongwith standard accessories. The contractor shall supply all fixing / supporting / accessories, devices, nuts, bolts, screws, hangers C.P. connector, elbows, extn. PSC & S.S. flanges as required and install the fixture / fitting in a neat workman like manner.				
	iii. No additional all fixing cost shall be paid for change in type of sanitary fixture or fitting.				
	iv. Provision of extension piece for final connection of CP fitting shall be supplied and installed by the contractor accordingly (as required).				
	v. Flaring should be done to C.P. Connectors before fixing it to Heaters & other fixtures.				
	vi. CP fittings and sanitary ware fittings should be fixed after acid washing is complete. All fittings shall be fixed with Teflon tape & white zinc (Safeda) only and not Hold tite or Super tite. Before fixing of Cp fittings, A) Cement grouting should be done around concealed elbows. (White cement in contractors scope) B) Cloths/duster should be used to fix CP fittings to avoid scratches/ damage etc. C) Water testing / cleaning shall be done of lines				
1	Fixing, testing and commissioning of white vitreous China wall mounting type European water closet with C.P. bolts, nuts, C.I. chair or other hanging arrangements, bakelite seat & cover with C.P. hinges & rubber buffers, C.P. brass screw, washer with all accessories. Including cutting & making good the walls, floors, slab wherever required.	Nos.	0.00	0.00	0.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
2	Fixing, testing and commissioning of Concealed Cistern for wall hung water closet with supporting arrangements, with proper line level and required civil works, complete with all accessories. Including cutting & making good the walls & floors wherever required.	Nos.	0.00	0.00	0.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
3	Fixing, testing and commissioning of white vitreous china floor mounting type European water closet with built-in exposed cistern of capacity 3 & 6 litres with C.P. bolts, nuts, supporting arrangements, bakelite seat & cover with C.P. hinges & rubber buffers, complete with all accessories. Including cutting & making good the walls & floors wherever required.	Nos.	396.00	0.00	396.00
4	Fixing, testing and commissioning of white vitreous china floor mounting type EWC or Anglo Indian water closet without cistern with C.P. bolts, nuts, supporting arrangements, bakelite seat & cover with C.P. hinges & rubber buffers, complete with all accessories. Including cutting & making good the walls & floors wherever required. (For Gr. Toilet)	Nos.	1.00	0.00	1.00
5	Fixing, testing and commissioning of exposed plastic cistern for WC, complete with all necessary fittings, flush pipes & accessories including all supports and making good. cistern dual flush, outlet from tank to w.c. (For Gr. Toilet)	Nos.	1.00	0.00	1.00
6	Fixing, testing & commissioning of Exposed Single lever wall mixer assembly with 150 mm long bend pipe OR hand shower with flexible pipe / shower rail and wall flanges complete as a full set as required.	Nos.	298.00	0.00	298.00
7	Fixing, testing & commissioning of Concealed Single lever wall diverter assembly with hand shower with flexible pipe, spout & hook for hand shower including O.H. Shower with arm C.P. wall flange complete as required.	Nos.	0.00	0.00	0.00
8	Fixing, testing and commissioning of exposed single lever sink tap wall or table mounted with flange complete with all necessary fittings, and making good.	Nos.	198.00	0.00	198.00
9	Fixing, testing & commissioning of ABS Rinsing spray / health faucet / Jet Plate etc. with regulator of approved quality with plastic flexible pipe 1.0 m long, wall hooked complete as required.	Nos.	397.00	0.00	397.00
10	Fixing, testing and commissioning of vitreous china below counter / above counter wash basin / semi counter wash basin with CP brass waste assembly, CP cast brass bottle trap with extension piece, wall flanges and rubber adapters for waste connection complete including CI / MS brackets duly painted, cutting and making good the walls floors wherever required	Nos.	0.00	0.00	0.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
11	Fixing, testing and commissioning of vitreous china wall hung flat back wash basin with C.I./M.S. brackets duly painted, C.P. brass waste, 32 mm diameter PP / C.P. cast brass bottle trap with extension piece and wall flange complete, including cutting and making good the walls/floors wherever required.	Nos.	397.00	0.00	397.00
12	Fixing, testing and commissioning of C.P. brass pillar cock “including providing & fixing connector” for pillar cock complete as required.	Nos.	397.00	0.00	397.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
13	Fixing, testing & commissioning of instant / storage type water heater of approved shade with copper / SS container, glasswool insulation, stove enamelled M.S jacket, thermostatically controlled inner heater with pilot neon lamps, suitable for working pressure of 5/ 8 kg/sqcm with suitable KW heater coil, 240 volts, single phase supply complete as required. <u>Including providing and fixing of g.i. brackets, anchor fastener, g.i. nut-bolt-washer etc. complete.</u> (Bathroom heater fixing above false ceiling)				
13.1	Capacity 3 litres (3KW)	Nos.	0.00	0.00	0.00
13.2	Capacity 3 litres (4.5KW)	Nos.	298.00	0.00	298.00
14	Fixing, testing and commissioning of C.P. brass Bib cock with C.P. wall flange, Washer etc. complete as required. (for Refuge flat / S. Toilet etc.)	Nos.	2.00	0.00	2.00
15	Fixing, testing and commissioning of C.P. brass angle valve with C.P. wall flange, Nut and Washer etc. complete as required. (for heater inlet, health faucet, exposed flush tank inlet, W/M inlet, Purifier inlet etc.) W/M & Purifier inlet Angle cock with coupling and plug.	Nos.	1488.00	0.00	1488.00
16	Providing, Fixing, testing and commissioning of S.S. 5" dia. round jali for floor traps etc. complete as required. (For refuge flats) 5" Round (Glossy finish) 2PC	Nos.	3.00	0.00	3.00
17	Providing, Fixing, testing and commissioning of 5" x 5" cockroach trap S.S. Jali for floor traps etc. complete as required. (for Kitchen sinks, pantry etc.) 5" x 5" Square (Glossy finish), 18 gauge, Cockroach Trap	Nos.	198.00	0.00	198.00
18	Providing, Fixing, testing and commissioning of 5" x 5" Square (Matt finish), 18 gauge 2PC (For Bathrooms, balcony / deck drain)	Nos.	596.00	0.00	596.00
	Note:-All accessories like screws for fixing sanitary ware or brackets, anchor fasteners, additional c.p. flanges, c.p. connectors, c.p. male threaded elbow etc. shall be supplied and fixed by contractor to complete the installation.				
	Scope also includes Testing & commissioning of all fixtures (CP/Sanitary) installed in sample flat, including maint. of same.				
TOTAL : "A" CARRIED TO SUMMARY					

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
B.	COLD & HOT WATER SUPPLY (INTERNAL)				
1	Concealed Piping (Hot & Cold water supply in Toilets and Kitchens)				
1.1	Providing, fixing, jointing and testing in position CPVC (Chlorinated Poly Vinyl Chloride) water supply pipes with fittings as per CTS SDR 11 (operating pressure - 7 Bar @ 82 Deg C and 28 Bar @ 23 Deg C).				
	Pipes shall be joined using solvent welded CPVC fittings i.e. Tees, Elbows, Couplers, Unions, Reducers, bushings etc. including transition fittings i.e. Brass Adaptors / Elbows (both Male & Female threaded) conforming to ASTM D-2846. ASTM F441 with only CPVC solvent cement conforming to ASTM F-493.				
	Cost shall be inclusive of (a) Making chasing in wall for the pipe, making good the same after piping and pressure testing by using 1:1 Geru mixed cement mortar including all Civil materials. or (b) Fixing/supporting the pipes (& fittings) at wall/ceiling level supported by split clamp with rubber lining type hangers etc, as per specification. PVC pipe sleeves of suitable higher size shall be provided wherever the pipes are crossing the walls / floors slab.				
	(c) Pressure testing of all concealed water supply pipes as per requirement (5 kg/cm2 for Gravity system & 10 kg/cm2 for Hydropneumatic system) (d) Curing of chasing finishing works for 7 days.				
	For hot & Cold water piping in toilets.				
	Nitrile rubber insulation for exposed Hot water pipes in ceiling or in shaft or on terrace. (15 to 25mm - 9mm thk. / 32 to 80mm = 13mm thk.)				
1.1.1	15 mm dia (CPVC pipes with fittings as per CTS SDR 11)	RM	4500.00	0.00	4500.00
1.1.2	20 mm dia (CPVC pipes with fittings as per CTS SDR 11)	RM	80.00	0.00	80.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
2	G.I. PIPING				
2.1	Providing, fixing, jointing and testing in position the following heavy class (Class-C) G.I. pipes conforming to IS:1239 cut to required lengths including all necessary fittings (All fittings shall confirm to IS:1879 - part 1 to 10) and specials such as bends, tees, unions, reducers, flanges & plugs etc. Fixing at wall/ceiling level supported by galvanized clamps, hangers etc, as per specification. Threading, jointing, and making proper connections.				
	For pump delivery lines only 'C' type Fabricated Hotdip galvanised brackets with PVC coated 'U' clamps shall be used for vertical and horizontal supports as per details given below				
	Vertical support spacing (Center to Center) 'C' type Fabricated Hotdip galvanised brackets with GI 'U' clamps - shall be as follows -15mm = 1800mm / 20mm & 25mm = 2000mm / 32mm to 50mm = 2800mm / 65mm & above = 3000mm Horizontal supports spacing (Center to Center) shall be as follows - 15mm = 1250mm / 20mm & 25mm = 2000mm / 32mm to 50mm = 2200mm / 65mm & above = 2500mm. Above are manufacturer's standards, but for vertical supports use 3 brackets / floor for residential building and 4 brackets / floor for commercial building or 1 m c/c whichever is less. For horizontal supports use 1.25 m c/c for all sizes.				
	For exposed piping for TMC, Pump Delivery-Suction, Tank Interconnection etc. O.H. Tank to Podium loop lines				
2.1.1	50 mm dia (For municipal connection)	RM	20.00	0.00	20.00
2.1.2	65 mm dia (Flushing delivery)	RM	150.00	0.00	150.00
2.1.3	80 mm dia (Domestic delivery)	RM	150.00	0.00	150.00
2.1.4	100 mm dia	RM	6.00	0.00	6.00
2.1.5	150 mm dia (Domestic & Flushing Suction, Tank interconnection)	RM	12.00	0.00	12.00
2.2	Providing, fixing, jointing and testing in position the following heavy class (Class-C) G.I. pipes conforming to IS:1239 cut to required lengths with Zoloto-M or 'R' brand GI fittings as per specifications and complete with fittings such as sockets, unions, reducers long or short bends, elbows, nipples flanges, checknuts, clamps etc.				
	Scope includes, Flanged connections with G.I. puddle sleeves provided in RCC walls with providing and fixing of required G.I. flanges, G.I. Nut-Bolt & Washers & rubber packing's.				
	Scope also includes Excavation of required width & depth, Pressure testing of piping and providing protection to embedded G.I. Pipes & fittings by wrapping hessian cloth over two coats of bitumen, & back filling of trench complete as required.				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
	For under ground piping for TMC including all Civil materials. (excavation, backfilling quantity considered separately)				
2.2.1	50 mm dia (For municipal connection)	RM	10.00	0.00	10.00
2.2.2	80 mm dia	RM	0.00	0.00	0.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
3	Providing, fixing, jointing and testing in position the following U-PVC pressure Pipes Sch. 40 as per ASTM D 1785 and plain end (pasted) . Cut to required lengths including all necessary fittings and specials such as bends, tees, unions, reducers, plugs etc in U-PVC pressure fittings Sch. 80 as per ASTM D 1785 Jointing with solvent cement. Use Cleaner, Primer and solvent for pipe dia. 50mm and more.				
	Scope includes, Flanged connections with G.I. puddle sleeves provided in RCC walls with providing and fixing of required G.I. flanges, G.I. Nut-Bolt & Washers & rubber packing's.				
	Scope also includes Pressure testing of piping, encasing of pipes with CC M-15 & back filling of trench complete as required.				
	For under ground piping for, Irrigation, STP, Sump pumps , Tanker inlet, DCB etc.				
	including all Civil materials. (excavation, concrete, backfilling quantity considered separately)				
3.1	50 mm dia (U.G. STP, TMC line, Irrigation line etc.)	RM	10.00	0.00	10.00
3.2	80 mm dia (U.G. Sump pump delivery, DCB, Tanker inlet, Irrigation line etc.)	RM	400.00	0.00	400.00
3.3	100 mm dia (U.G. Tanker inlet / Sump pump delivery / STP line)	RM	10.00	0.00	10.00
4	Providing, fixing, jointing and testing in position the following U-PVC pressure Pipes Sch. 40 as per ASTM D 1785 and plain end (pasted) . Cut to required lengths including all necessary fittings and specials such as bends, tees, unions, reducers, plugs etc in U-PVC pressure fittings Sch. 80 as per ASTM D 1785 Jointing with solvent cement. <u>Including G.I. Brackets with PVC coated 'U' clamps</u> fixing at wall/ ceiling level, hangers etc. as per specification. Cutting hole in wall and making good the same with cement concrete M-15 complete as required. pvc pipe sleeve of larger diameter shall be provided wherever the pipes crossing the walls / floor slab and sealing the sleeves as per consultants requirement. Use Cleaner, Primer and solvent for pipe dia. 50mm and more.				
	Brass insert male adaptor shall be used for toilet / kit. inlet to branch pipe connection (15mm & 20mm). For terrace loop branch valves, metal insert adaptors shall be used.				
	For suspended piping (other than loop) GI split clamp with rubber lining with rod/nut/washer etc. shall be used as per required spacing. <u>Every 5 mtr. Interval G.I. bracket {including painting} with pvc coated clamps shall be used.</u>				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
	Scope also includes miscellaneous works like – U.G. & O.H. tank overflow, drain {Fire/Flush/Domestic} , 25 dia. loop air piping with supports, mosquito nets etc.				
	for vertical supports use 3 brackets / floor for residential building and 4 brackets / floor for commercial building or 1 m c/c whichever is less. Additional supports shall be provided for water supply branch pipes in shaft, which are more than 500mm long. (2 supports, if branch is more than 1000mm long)				
	For vertical lines, Terrace loop & branch pipes, 'C' type Fabricated Hotdip galvanised brackets with PVC coated 'U' clamps shall be used.				
	Horizontal supports spacing (Center to Center) shall be as follows - 15mm and 20mm = 500mm / 25mm = 600mm / 32mm = 700mm / 40mm = 800mm / 50mm = 900mm and 65mm and bigger dia. = 1000mm.				
	(For External water supply downtake / Uptake piping in shaft, staircase servant toilet open piping, Terrace loop, STP line, DCB line, tanker line, sump pump lines etc.)				
4.1	15 mm dia	RM	1470.00	0.00	1470.00
4.2	20 mm dia	RM	675.00	0.00	675.00
4.3	25 mm dia	RM	1400.00	0.00	1400.00
4.4	32 mm dia	RM	1350.00	0.00	1350.00
4.5	40 mm dia	RM	800.00	0.00	800.00
4.6	50 mm dia (Terrace loop, tank overflow, tank drain, STP line etc.)	RM	663.00	0.00	663.00
4.7	65 mm dia (Terrace loop)	RM	210.00	0.00	210.00
4.8	80 mm dia (Exposed Sump pump delivery, Terrace loop, DCB, Tanker inlet, Irrigation line, STP line etc.)	RM	472.00	0.00	472.00
4.9	100 mm dia (Exposed Sump pump delivery, Terrace loop, Tanker inlet, STP line etc.)	RM	264.00	0.00	264.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
5	Providing, fixing, jointing and testing in position the following U-PVC pressure Pipes Sch. 80 as per ASTM D 1785 and plain end (pasted) . Cut to required lengths including all necessary fittings and specials such as bends, tees, unions, reducers, plugs etc in U-PVC pressure fittings Sch. 80 as per ASTM D 1785 Jointing with solvent cement. Including G.I. Brackets with PVC coated 'U' <u>clamps</u> fixing at wall/ ceiling level, hangers etc. as per specification. Cutting hole in wall and making good the same with cement concrete M-15 complete as required. pvc pipe sleeve of larger diameter shall be provided wherever the pipes crossing the walls / floor slab and sealing the sleeves as per consultants requirement. Use Cleaner, Primer and solvent for pipe dia. 50mm and more.				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
	Brass insert male adaptor shall be used for toilet / kit. inlet to branch pipe connection (15mm & 20mm). For terrace loop branch valves, metal insert adaptors shall be used.				
	For suspended piping (other than loop) GI split clamp with rubber lining with rod/nut/washer etc. shall be used as per required spacing. <u>Every 5 mtr. Interval G.I. bracket {including painting} with pvc coated clamps shall be used.</u>				
	Scope also includes miscellaneous works like – U.G. & O.H. tank overflow, drain {Fire/Flush/Domestic} , 25 dia. loop air piping with supports, mosquito nets etc.				
	for vertical supports use 3 brackets / floor for residential building and 4 brackets / floor for commercial building or 1 m c/c whichever is less. Additional supports shall be provided for water supply branch pipes in shaft, which are more than 500mm long. (2 supports, if branch is more than 1000mm long)				
	For vertical lines & branch pipes, 'C' type Fabricated Hotdip galvanised brackets with PVC coated 'U' clamps shall be used.				
	Horizontal supports spacing (Center to Center) shall be as follows - 15mm and 20mm = 500mm / 25mm = 600mm / 32mm = 700mm / 40mm = 800mm / 50mm = 900mm and 65mm and bigger dia. = 1000mm.				
	(For External water supply downtake piping in shaft for 50mm & bigger dia.)				
5.1	50 mm dia (sch. 80)	RM	0.00	0.00	0.00
5.2	65 mm dia (sch. 80)	RM	0.00	0.00	0.00
5.3	80 mm dia (sch. 80)	RM	0.00	0.00	0.00
5.4	100 mm dia (sch. 80)	RM	0.00	0.00	0.00
6	Providing & fixing full way lever operated forged brass ball valve of brass body with forged brass hard chromeplated steel ball tested to a pressure not less than 20 Kg / sqcm with threaded / flanged joints complete with nuts, bolts, gaskets, washers etc.				
	For all Water supply branch on terrace / podium level and in shafts, Terrace loop, Hydropneumatic loop, U.G. & O.H. tank drain {Fire/Flush/Domestic}, pump priming, before & after PRV, below air valve, Municipal etc.				
	PVC (male / female) adaptors can be used for Ball Valves of water supply downtake branch pipes in toilet & kitchen shaft. Union shall be provided near all ball valves.				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
	Brass Insert (male / female) adaptors shall be used for Ball Valves of before & after PRV. Union shall be provided near all ball valves.				
6.1	15 mm dia	Nos.	960.00	0.00	960.00
6.2	20 mm dia	Nos.	50.00	0.00	50.00
6.3	25 mm dia	Nos.	228.00	0.00	228.00
6.4	32 mm dia	Nos.	174.00	0.00	174.00
6.5	40 mm dia	Nos.	18.00	0.00	18.00
6.6	50 mm dia	Nos.	31.00	0.00	31.00
6.7	65 mm dia (For loop)	Nos.	2.00	0.00	2.00
6.8	80 mm dia (For loop)	Nos.	2.00	0.00	2.00
7	Providing & fixing butterfly valve (upto 150mm dia with hand lever operation & above with gear box operation). Tested to a pressure not less than as mentioned below, Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers.				
	For tank interconnection, pump suction / delivery, STP, bore well, Tanker inlet, Sump Pump, RWH lines etc.				
7.1	65 mm dia PN-25 gear operated.	Nos.	0.00	0.00	0.00
7.2	65 mm dia (Main Flushing pump) PN-16	Nos.	5.00	0.00	5.00
7.3	65 mm dia (OHT) PN-10	Nos.	0.00	0.00	0.00
7.4	100 mm dia (Main Domestic, Flushing pump) PN-16	Nos.	0.00	0.00	0.00
7.5	80 mm dia (DCB, Sump Pump, Irrigation line, Tanker line, STP line, Tanker line, OHT) PN-10.	Nos.	17.00	0.00	17.00
7.6	80 mm dia (Main Domestic pump) PN-16.	Nos.	5.00	0.00	5.00
7.7	100 mm dia (Suction header, Sump Pump, Irrigation pump, Loop, OHT, Tanker line) PN-10.	Nos.	9.00	0.00	9.00
7.8	150 mm dia (Tank outlet, Suction header, Interconnection etc.) PN-10.	Nos.	2.00	0.00	2.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
8	Providing, fixing, testing and commissioning of PRV in water supply downtake / branch pipes. To be installed inside shaft, on-line type. Additional supports shall be provided to PRV as per site condition. Brass insert (male / female) adaptors shall be used for all PRV's. Also 'Tee' used before & after PRV for pressure gauge shall be brass insert type.				
8.1	15 mm dia	Nos.	0.00	0.00	0.00
8.2	20 mm dia	Nos.	0.00	0.00	0.00
8.3	25 mm dia	Nos.	92.00	0.00	92.00
8.4	32 mm dia	Nos.	68.00	0.00	68.00
8.5	40 mm dia	Nos.	0.00	0.00	0.00
8.6	50 mm dia	Nos.	0.00	0.00	0.00
9	Providing, fixing, testing and commissioning water meter with direct reading dial in KL with all integral parts of gun metal or brass with necessary fitting such as threaded pieces, unions pressure gauge and isolation cock, flanges piece for future removal, flanges / unions complete with all necessary testing charges and obtaining test certificates from municipal authorities, on following size pipe lines. (Municipal water , STP, Bore well & DCB lines). (ball valve & strainer considered in respective item).				
9.1	25 mm dia	Nos.	1.00	0.00	1.00
9.2	50 mm dia	Nos.	2.00	0.00	2.00
9.3	80mm dia. (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (STP, DCB line if required).	Nos.	1.00	0.00	1.00
10	Providing & fixing Auto Air vent for cold water supply risers, suitable for pressure not less than 15 Kg/Sq.cm. (For top 3 floor loop).				
10.1	15 mm dia	Nos.	8.00	0.00	8.00
10.2	25 mm dia	Nos.	0.00	0.00	0.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
11	Providing & fixing of heavy quality Non Return valves of gun metal / brass / CI body with screwed / flanged ends pressure as mentioned below & as per details mentioned in approved materials list etc. complete (For plant room, sump pump, etc.) Flanged end - Dual plate type.				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
11.1	65 mm dia (Main Flushing pump) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-16) .	Nos.	3.00	0.00	3.00
11.2	100 mm dia (Main Domestic pump) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-16) .	Nos.	0.00	0.00	0.00
11.3	80 mm dia (DCB / sump pump / Tanker / STP) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-10) .	Nos.	2.00	0.00	2.00
11.4	80 mm dia (Main Domestic pump)) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-16) .	Nos.	3.00	0.00	3.00
11.5	100 mm dia (Irrigation / sump pump) (Flanged End) Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers (PN-10)	Nos.	2.00	0.00	2.00
12	Providing & fixing of heavy quality Strainers (with 400 micron SS mesh) of gun metal / brass / CI body with screwed / flanged ends 16kg class as per details mentioned in approved materials list etc. complete (For Terrace Loop, Water meter etc.) . CI body Strainers with anti corrosive coating.				
	Including rubber gasket, G.I. flanges, G.I. nuts, G.I. bolts, G.I. washers for flanged end strainers.				
12.1	25 mm dia. (for water meter) (GM / Brass - Y Type strainer)	Nos.	1.00	0.00	1.00
12.2	50 mm dia (for water meter) (GM / Brass - Y Type strainer)	Nos.	2.00	0.00	2.00
12.3	50 mm dia (for loop / pump suction) (VARIE - ECS B).	Nos.	0.00	0.00	0.00
12.4	65 mm dia (terrace loop) (VARIE - ECS B).	Nos.	2.00	0.00	2.00
12.5	80 mm dia (terrace loop) (VARIE - ECS B).	Nos.	2.00	0.00	2.00
12.6	65 mm dia (for water meter, pump suction - flange type).	Nos.	2.00	0.00	2.00
12.7	80 mm dia (for water meter, pump suction - flange type).	Nos.	3.00	0.00	3.00
12.8	100 mm dia (for terrace loop / pump suction / water meter - flange type)	Nos.	5.00	0.00	5.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
13	Providing and fixing 600 mm dia cast iron (heavy duty) water tank circular cover with frame 40 mm CI seat & lockable arrangement complete in all regards (Total WT of cover and frame to be not less than 20.8 kg). Including painting with 3 coats of bitumastic paint. (All UG & OH tanks).	Nos.	15.00	0.00	15.00
14	Providing & Constructing of block masonry chamber (for housing valves, Meters etc.) of block work with cement mortar 1:5 (1 cement:5 fine sand) plastering internal face with cement mortar 1:3 (1 cement: 3 fine sand and rough plaster on outer face with a floating coat of neat cement with required duty Ductile Iron manhole cover and frame with locking arrangement, Including excavation, dewatering, refilling, watering, ramming and removing the surplus excavated material complete as required. (All sizes are clear internal sizes).				
14.1	Size 900 x 600 x 900 mm deep with 900 x 600 Heavy duty D.I. frame and cover. (104kg / C-250 / MR 17)	Nos.	2.00	0.00	2.00
14.2	Size 900 x 450 x 900 mm deep with 900 x 450 Medium duty D.I. frame and cover. (67kg / B-125 / MR 18)	Nos.	0.00	0.00	0.00
15	Providing & fixing of vapour vent pipes of 50mm with GI Tee, Elbow, Pvc Mosquito proof jali including required jointing material etc. complete. (1 set = 2" pipex300mm + 1 Coupling + 1 Tee + 2 Elbow + 2 mosquito jali).				
15.1	50 mm dia	Set	24.00	0.00	24.00
16	Providing, fixing, jointing and testing in position approved quality of dial type pressure gauge with glycerin filled S.S. Bourdon tube pressure scale of 0 to 10 / 15 / 25 kg / cm ² . Application areas (activity) - A) Pump room delivery lines (flushing / domestic) B) Downtake PRV setting (4 gauges min.) C) For Terrace Booster system.				
	Calibration shall be done for pressure gauges and certificate shall be submitted before starting of pressure testing works for any plumbing lines and for pressure gauges for pump rooms. (as per requirement of ISO) With below details - a) Make of gauge - b) Pressure Range of gauge - c) Date of calibration -	Nos.	12.00	0.00	12.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
17	Providing, fixing, jointing and testing in position double flanged flexicon rubber expansion joint <u>for Domestic & Flushing pump delivery lines + between third NRV & Butterfly valve</u> (suitable for system test pressure) of standard length as per manufacturers specs including rubber gaskets, GI flanges, GI nuts, bolts and washers complete as required. PN-16				
17.1	65mm dia.	Nos.	3.00	0.00	3.00
17.2	80mm dia.	Nos.	3.00	0.00	3.00
17.3	100mm dia. (PN-10, for irrigation pump)	Nos.	2.00	0.00	2.00
18	Providing & fixing of heavy quality Ball type Non Return Valve of CI body with flanged ends 10kg class, Nitrile rubber coated ball with inside lining of Rubber for valve. Including rubber gasket, G.I. flanges, GI nuts, GI bolts, GI washers. (For Flushing Tanker inlet / Sewage pump delivery) .				
18.1	80mm dia.	Nos.	0.00	0.00	0.00
18.2	100mm dia.	Nos.	0.00	0.00	0.00
TOTAL : "B" CARRIED TO SUMMARY					

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
C.	INTERNAL DRAINAGE (SOIL, WASTE & VENT)				
1	Providing, fixing, jointing, testing and commissioning of UPVC pipe Type-A for R.W. upto 160mm dia. confirming to IS 13592 & AGRI for R.W. above 160 mm dia confirming to IS:4985 cut to required lengths including all necessary fittings and specials such as 45 or 90 deg. bends, Tee's, Y's, access pipe with solvent cement jointing. Fixing at wall / ceiling level supported by galvanized brackets & pvc coated clamps & hangers etc.				
	All pipe diversions shall be done with 45 degree bends where ever required due to beam offsets or cutout changes at any level. For vertical support, fix 4 brackets/floor or 1 m c/c whichever is less.				
	For Vertical Rain Water Pipes				
	Scope includes connection of service floor drains to Multiples RWP pipes with using Y' & Bends.				
1.1	160 dia (Type A) Pasted type.	RM	325.00	0.00	325.00
1.2	110 dia (Type A) Pasted type.	RM	350.00	0.00	350.00
1.3	75 dia (Type A) Pasted type (for small terraces of watchmen cabin or other structure etc. as per site condition).	RM	10.00	0.00	10.00
2	Providing, fixing, jointing, testing and commissioning of UPVC pipe Type-A for vent & Type-B for Soil, Waste & A.C. drains upto 160mm dia. confirming to IS 13592 cut to required lengths including all necessary fittings and specials such as as 45 or 90 deg. bends, Swept Tee's , Y's, access pipe with rubber ring jointing. PVC Cowl on top with solvent joint.				
	Fixing at wall / ceiling level supported by galvanized brackets & PVC coated clamps & hangers etc. Vertical stacks shall be truly vertical & parallel to the wall and supported on M.S. Hot dip Galvanised brackets with PVC coated 'U' clamps, so that the pipes are at least 100mm away from the finished surface/ plaster. Also 75mm to 100mm gap between 2 pipes should be maintained. All pipe diversions shall be done with 45 degree bends where ever required due to beam offsets or cutout changes at any level.				
	For vertical support, fix 3 brackets/floor or 1 m c/c whichever is less.				
	For Vertical Soil, Waste, Vent, Anti Siphonage, AC Drain etc.				
	Only Swept Tee with Door shall be used for all Soil and Waste lines which are with Tee connections shown on drawing.				
2.1	160 dia (Type B)	RM	0.00	0.00	0.00
2.2	110 dia (Type B) (Soil pipe).	RM	1284.00	0.00	1284.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
2.3	75 dia (Type A) (Toilet - Anti Siphonage main & branch pipe / Vent for starting point).	RM	1545.00	0.00	1545.00
2.4	75 dia (Type B) (Toilet / Kitchen Waste pipe).	RM	1926.00	0.00	1926.00
2.5	75 dia (Type B) (Balcony drain pipe).	RM	642.00	0.00	642.00
2.6	75 dia (Type A) (AC Drain pipe). (With provision of single Y + 45 deg. Bend on every floor)	RM	642.00	0.00	642.00
3	Providing and fixing including labour for fixing in position PVC Floor trap with 75mm dia outlet and a trap seal of 50mm for kitchen and bath wet area & 20 to 25mm for dry area. If two or three floor trap connections are with 'Y' or separately connected with waste lines, then all floor traps shall be of 50mm water seal. Making proper connection, cutting chase / hole in floors /slabs and bringing the same in proper condition and shape after placing the trap in right position complete as required with encasing in CC M-15.				
3.1	100 mm inlet and 75 mm outlet. (UPVC) 50mm water seal for toilet dry / wet area of 250mm sunk toilets.	Nos.	17.00	0.00	17.00
3.2	100 mm inlet and 75 mm outlet. (UPVC) 50mm water seal for toilet wet area of 50mm sunk toilets (suspended with bottom support).	Nos.	382.00	0.00	382.00
3.3	100 mm inlet and 75 mm outlet. (UPVC) 50mm water seal for kitchen or pantry area.	Nos.	199.00	0.00	199.00
3.4	100 mm inlet and 75 mm outlet. (UPVC) Regular water seal for balcony drain.	Nos.	199.00	0.00	199.00
4	Providing, fixing, jointing, testing and commissioning of UPVC pipe Type-B for Soil, Waste upto 160mm dia. confirming to IS 13592 cut to required lengths including all necessary fittings and specials such as as 45 or 90 deg. bends, Tee's, Y's, access pipe with solvent cement jointing. (For Pipes buried in flooring).				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
	Setting the pipes in cement concrete M-15, 100mm all around of pipes as per specifications after testing of pipes. SWR range pasted pipes (Type-B) & fittings (without rubber rings) with solvent joints for all sunken area (concealed) piping for toilets / kitchens (Water testing of pipes to be done after 12 hours, after last solvent cement joint. Where ever possible use Aqua Drain test plugs for testing of lines.) Including civil material for pcc+encasing (concrete quantity considered separately).				
4.1	110 dia (Type B) (Toilets / Balcony).	RM	90.00	0.00	90.00
4.2	110 dia (Type B) (Kit.).	RM	70.00	0.00	70.00
4.3	75 dia (Type B) (Toilets).	RM	10.00	0.00	10.00
4.4	75 dia (Type B) (Kit. & Balcony).	RM	200.00	0.00	200.00
5	Providing, fixing, jointing, testing and commissioning of UPVC pipe Agri. range 6 kgf pipes and fittings for 40, 50 & 63mm dia. cut to required lengths including all necessary fittings and specials such as as 45 or 90 deg. bends, Tee's, Y's, access pipe with solvent cement jointing. Vertical pipes (above sunk level) in necessary wall chases and finishing of same with geru mixed cement mortar. Including civil material for pcc+encasing (concrete quantity considered separately).				
5.1	Setting the pipes in cement concrete M-15, 100mm all around of pipes as per specifications after testing of pipes. (Water testing of pipes to be done after 12 hours, after last solvent cement joint. Where ever possible use Aqua Drain test plugs for testing of lines.)				
	For Pipes Buried in Toilets / kit. from Wash Basin, Sink, etc to Floor trap / Main Drain (50mmx25mm reducer socket and 40mmx25mm reducer socket shall be fixed for washing machine / dish washer outlet).				
5.1.1	40 mm dia (Toilets).	RM	40.00	0.00	40.00
5.1.2	40 mm dia (Kitchens).	RM	400.00	0.00	400.00
5.1.3	50 mm dia (Toilets).	RM	0.00	0.00	0.00
5.1.4	50 mm dia (Kitchens).	RM	10.00	0.00	10.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
5.2	Providing and fixing including labour for fixing UPVC pasted pipes (Type-B) & fittings (without rubber rings) with solvent joints of 75 & 110mm dia. and Agri. range 6 kgf pipes and fittings for 40, 50 & 63mm dia. for all suspended horizontal piping for toilets / kitchens including all necessary fittings and specials such as as 45 or 90 deg. bends, Tee's, Y's, access pipe etc.				
	Pipe portion going in concrete area shall be applied with solvent and fine sand. (Water testing of pipes to be done after 12 hours, after last solvent cement joint. Where ever possible use Aqua Drain test plugs for testing of lines.)				
	Supports shall be Split clamp with rubber lining for entire suspended pipings as per specs. Split clamp rod & Anchor bolt dia. – 8mm for 40mm & 50mm pipes and 10mm for 63mm, 75mm and 110mm pipes. Additional supports to be provided near fittings if required as per site condition & instructions of engineer in-charge. Horizontal supports spacing (Center to Center) shall be as follows - 40mm = 400mm / 50mm= 500mm / 75mm= 750mm / 110mm = 1000mm / 160mm & above = 1500mm.				
	For suspended pipes in Toilets / kitchens from Wash Basin, Sink, Washing Machine, Dish Washer etc to 'P' Trap. Supports required at bottom of 'P' Trap & Floor Trap.				
	Vertical pipes (above slab level) in necessary wall chases and finishing of same with geru mixed cement mortar.				
	Horizontal SWR drains for 50 / 65mm sunk toilets and Kitchens (suspended piping) (50mm x 25mm reducer socket and 40mm x 25mm reducer socket shall be fixed for washing machine / dish washer outlet).				
5.2.1	40 mm dia	RM	650.00	0.00	650.00
5.2.2	50 mm dia	RM	0.00	0.00	0.00
5.2.3	63 mm dia	RM	0.00	0.00	0.00
5.2.4	75 mm dia	RM	185.00	0.00	185.00
5.2.5	110 mm dia	RM	660.00	0.00	660.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
6	Providing, fixing, Testing and Commissioning uPVC agricultural pipe with necessary fittings, GI brackets, clamps, split clamp with rubber etc. complete. Concealed / false ceiling area pipes if any, with elastomeric rubber insulation (clamps shall be fixed after insulation) and exposed area pipes without insulation & supports at 1 mtr. c/c. (For A.C. Drain connection in shaft, 'U' trap to be done with using fittings before connecting to vertical pipe) Quantity to be check & order as per site condition.				
6.1	25 mm dia (8kgf or 10kgf). (For 250mm sunk area sleeve)	RM	170.00	0.00	170.00
6.2	20 mm dia (8kgf or 10kgf) .	RM	0.00	0.00	0.00
7	Providing, fixing, jointing, testing and commissioning of heavy quality approved make CI spigot and socket pipes for soil, waste and vent, conforming to IS 3989 including all fittings like offsets, 45 deg. bends, single or double junctions of different degrees, with access doors, etc. including providing & fixing with caulking spun yarn and DRIP seal jointing with testing of pipes and making the joints leak proof. including fixing them vertically against walls on M.S. Hot dip Galvanised brackets & G.I. 'U' clamps or hangers including all scaffolding, testing after laying the pipes as specified etc. Complete.				
	Vertical SWR drains in shafts for toilets / kit. (ground to basement levels) with C.I. Cleaning pc. at bottom. (where applicable).				
7.1	80 mm dia	RM	0.00	0.00	0.00
7.2	100 mm dia (including canopy drain).	RM	8.00	0.00	8.00
8	Providing and fixing P.V.C. WC connector with rubber lip ring to suit WC outlet and proper connection complete as required .				
8.1	For Servants / drivers toilets (Regular).	Nos.	1.00	0.00	1.00
9	Providing and fixing including labour for fixing in position PVC 'P' (110mm x 75mm) Trap with necessary support at bottom. Connection with pvc puddle sleeve provided with solvent joint etc. complete.				
	For toilet shaft drains	Nos.	1.00	0.00	1.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
10	Providing, fixing, Testing and Commissioning of uPVC Puddle Flanges (Type-B pipe only) including fixing with BCP screws (35 x 8), 4 nos. per flange. Center of flange shall be as per w.c. outlet & 'P' Trap outlet dimensions and same shall be in plumb before fixing screws.				
10.1	110mm dia x 600mm long (For w.c. outlet, floor drain & 'P' Trap etc.) .	Nos.	764.00	0.00	764.00
10.2	50mm dia x 600mm long (For dish washer / W.M. outlet etc.)	Nos.	0.00	0.00	0.00
10.3	40mm dia x 800mm long (For basin outlet / sink outlet / W.M. outlet etc.)	Nos.	382.00	0.00	382.00
11	Providing, Fixing, Testing & Commissioning of C.I. rain water Grating.				
11.1	100mm dia. Round (dome shape) for 75 mm outlet.	Nos.	0.00	0.00	0.00
11.2	150mm dia. Round (dome shape) for 110 mm outlet.	Nos.	4.00	0.00	4.00
11.3	225mm dia. Round (dome shape) for 160 mm outlet .	Nos.	0.00	0.00	0.00
11.4	200 x 200 square for 160 mm outlet.	Nos.	8.00	0.00	8.00
11.5	150 x 150 square for 110 mm outlet.	Nos.	8.00	0.00	8.00
11.6	300 x 300 square for 250 mm outlet.	Nos.	0.00	0.00	0.00
TOTAL : "C" CARRIED TO SUMMARY					

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
D.	EXTERNAL SEWAGE / RAIN WATER DRAINAGE				
1	Excavation in trenches for pipelines including trimming bottom and sides, cutting for sockets, ramming of bottom and disposal of surplus earth etc.				
1.1	Depth of Excavation not exceeding 1.5m (Excavation in all classes of soil).	Cum	250.00	0.00	250.00
1.2	Extra over 1.1 above for depths exceeding 1.5m but not greater than 2.5m (Excavation in all classes of soil).	Cum	110.00	0.00	110.00
1.3	Excavation in slush.	Cum	0.00	0.00	0.00
1.4	Excavation in hard rock / existing concrete. (refilling in 200mm layers with selected excavated soil)	Cum	0.00	0.00	0.00
1.5	Providing and laying cement concrete M-15 as per specifications for PCC / Encasing / Foundation etc. (including 250mm sunk toilet / kitchen / balcony pipe encasing)	Cum	80.00	0.00	80.00
1.6	Backfilling of trenches with murum / soil.	Cum	200.00	0.00	200.00
2	Providing, fixing, jointing and testing in position the following UPVC pipes confirming to IS: 4985 of 6 Kg / Sq.cm for drainage system cut to required length. Pipe to be laid below ground level in trenches upto required depth, dewatering, jointing of pipes with solvent cement, water testing of joints after 1 hour and setting the pipes in cement concrete M-15 as per specifications 150mm thk. all around, refilling of trenches, and making good the same complete as required.				
	Scope of work includes piping and manholes upto existing manhole or municipal manhole on road and connection with breaking the wall of municipal / road manhole and making good the same with Cement mortar 1:3, finished with a floating coat of neat cement and making the same good etc. complete.				
	For RWP System from Plumbing duct to RW chamber, Drain chamber Branch pipes.				
	Including civil material for pcc+encasing (concrete quantity considered separately).				
2.1	200 mm dia	RM	0.00	0.00	0.00
2.2	160 mm dia	RM	25.00	0.00	25.00
2.3	110 mm dia	RM	60.00	0.00	60.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
2.4	75 mm dia	RM	20.00	0.00	20.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
3	Providing, fixing, jointing and testing in position the following Double wall corrugated H.D.P.E. pipes, for non pressure storm water and sewerage applications. (As per ISO 21138 – 3) for drainage system cut to required length.				
	Pipe to be laid below ground level in trenches upto required depth, dewatering, jointing of pipes with solvent cement, water testing of joints after 1 hour and setting the pipes in cement concrete M-15 as per specifications 150mm thk. all around, refilling of trenches, and making good the same complete as required.				
	Scope of work includes piping and manholes upto existing manhole or municipal manhole on road and connection with breaking the wall of municipal / road manhole and making good the same with Cement mortar 1:3, finished with a floating coat of neat cement and making the same good etc. complete.				
	For RWP System from building to SWD, RWH Piping, Building Drainage etc. - Main and Branch pipes.				
	DWC SN4 pipes shall be used up to 1.5mtr. deep drain lines and SN8 pipes shall be used if depth is more than 1.5 mtr.				
	Including civil material for pcc+encasing (concrete quantity considered separately).				
3.1	100 mm Int. dia (SN-4)	RM	0.00	0.00	0.00
3.2	150 mm Int. dia (SN-8)	RM	80.00	0.00	80.00
3.3	200 mm Int. dia (SN-8)	RM	80.00	0.00	80.00
3.4	250 mm Int. dia (SN-8)	RM	20.00	0.00	20.00
3.5	300 mm Int. dia (SN-8)	RM	25.00	0.00	25.00
4	Providing and fixing round mouthed pvc Bottle gully trap with 300mm x 300mm chamber with CI cover at top including embedding the trap in cement concrete M-15 construction of brick masonry chamber internal clear size 300 x 300 mm in brick masonry 250mm thick in cement sand mortar 1:4 over M-15 cement concrete bid 150mm thick, plastering inside outside and all exposed surfaces with 1:3 cement sand plaster 15mm thick.				
	Including providing and fixing of CI frame & cover size 300 x 300mm weighing together not less than 7.8Kgms painting all CI items with 3 coats of approved bituminastic paint etc. complete.				
	For all waste lines including AC / Balcony / Shaft drain lines etc.	Nos.	21.00	0.00	21.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
5	Providing & Constructing block masonry inspection chamber of below sizes and depth upto 1375mm including the following: a) 200 thick block masonry walls in CM 1:4 b) C.C. foundation bed in M-15, 150 mm thick size 1600 x 1150 c) Cement plaster inside and outside in CM 1:3 (with cement ghotai from inside). d) CC M-15 in haunches and channels finished smooth in CM 1:2 e) C C M-15 cap on top 200 thick f) Providing & fixing of Ductile / Cast iron cover & frame g) 75 wide fillet (wata) all round block / brick masonry and c.c. bed in CM 1:3 h) All excavation in any type of soil removal and disposing off surplus earth after backfilling as directed including dewatering.				
	All civil materials in contractors scope.				
5.1	900mm x 450mm				
5.1.1	with 80 kg D.I. Cover (Item code no = C-250 / MR 14).	Nos.	9.00	0.00	9.00
5.1.2	with 67 kg D.I. Cover (Item code no = B-125 / MR 18).	Nos.	0.00	0.00	0.00
5.1.3	with 50 kg D.I. Cover (Item code no = A-15 / MR 4).	Nos.	0.00	0.00	0.00
5.2	900mm x 600mm				
5.2.1	with 104 kg D.I. Cover (Item code no = C-250 / MR 17).	Nos.	2.00	0.00	2.00
5.2.2	with 84 kg D.I. Cover (Item code no = B-125 / MR 19).	Nos.	0.00	0.00	0.00
5.2.3	with 65 kg D.I. Cover (Item code no = A-15 / MR 7).	Nos.	0.00	0.00	0.00
5.3	600mm x 600mm				
5.3.1	with 62 kg D.I. Cover (Item code no = C-250 / MS 24).	Nos.	8.00	0.00	8.00
5.3.2	with 58 kg D.I. Cover (Item code no = B-125 / MS 23).	Nos.	0.00	0.00	0.00
5.3.3	with 40 kg D.I. Cover (Item code no = A-15 / MS 6).	Nos.	0.00	0.00	0.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
6	Providing & Constructing brick masonry conical manholes including the following: a) Bed PCC for chamber to be M-15 concrete 200 mm thick, with 6" projection. b) Chamber to be constructed using bricks 225mm thick wall. c) Providing & fixing of Ductile / Cast iron cover & frame d) Plastic steps to be provided. (Copolymer Polypropylene, Steel reinforced manhole step, is injection molded around a e) Size of chamber - 1200 mm (for depths > 1375mm and < 2285mm), or 1500 mm (for depth > 2285mm) at bottom, and 0.53 m. at top plaster 1:1 (out) and 1:1 (inside) block masonry 1:3 f) Haunching & benching M-15. g) Depth beyond 2285mm should be in double block 200mm cylindrical. h) C C M-15 cap on top 300 thick i) Cement plaster inside and outside in CM 1:2 (with cement chetai on both sides) All civil materials in contractors scope.				
6.1	Depth upto 2285mm (1200mm dia. Clear at bottom).				
6.1.1	with 60 kg D.I. Cover (Item code no = C-250 / MC 25) 600mm dia.	Nos.	8.00	0.00	8.00
6.1.2	with 53 kg D.I. Cover (Item code no = B-125 / MC 24) 600mm dia.	Nos.	0.00	0.00	0.00
6.1.3	with 42 kg D.I. Cover (Item code no = A-15 / MC 12) 600mm dia.	Nos.	0.00	0.00	0.00
6.2	Depth beyond 2285mm (1500mm dia. Clear at bottom) in double block 200mm cylindrical.				
6.2.1	with 60 kg D.I. Cover (Item code no = C-250 / MC 25) 600mm dia.	Nos.	0.00	0.00	0.00
6.2.2	with 53 kg D.I. Cover (Item code no = B-125 / MC 24) 600mm dia.	Nos.	0.00	0.00	0.00
6.2.3	with 42 kg D.I. Cover (Item code no = A-15 / MC 12) 600mm dia.	Nos.	0.00	0.00	0.00
7	Providing, Fixing, testing & commissioning of Nu-Drain UG Ultra Inspection chambers of Supreme including the following: a) Bed PCC for chamber to be M-15 concrete 100 mm thick, with 6" projection. b) Encasing of Inspection Chamber / riser pipes (100mm all around) with CC M-15 All civil materials in contractors scope.				
7.1	Left hand 45 deg. and 90 deg. Junction (315mm x 110mm x 110mm x 110mm) including below items.	Nos.	2.00	0.00	2.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
	a) 315mm dia. x 154mm ht. Riser with rubber seal (2 or 3 nos. as per depth)				
	b) 315mm x 315mm Frame & Cover (Heavy duty)				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
7.2	Right hand 45 deg. and 90 deg. Junction (315mm x 110mm x 110mm x 110mm) including below items.	Nos.	3.00	0.00	3.00
	a) 315mm dia. x 154mm ht. Riser with rubber seal (2 or 3 nos. as per depth)				
	b) 315mm x 315mm Frame & Cover (Heavy duty)				
8	Providing, Fixing, Testing, & commissioning of Hubless centrifugally cast iron pipes and fittings (Epoxy coated inside & outside) as per ISO-6594 / IS-15905 Standard and jointing with ASTM C1277 marked stainless steel shielded coupling with a double stainless steel bolt & Screw Housing Incorporating a elastometric rubber gasket , system having noise level certification of not more than 25db(A) at 2L/S & inclusive of all necessary specials like bends, tees, offsets, junctions, cowl end plug, inspection pipe, etc. laid under floor/fixed on walls and in pipe shafts. The rate shall be including angle supports, hanger, nuts bolts, anchor fasteners, fixing clamps/ channels with U-bolts , (spacing and details mentioned below) etc. complete. Cutting, chases/holes in floors / walls / slab. (For Soil & Waste Water Pipes upto 300mm Dia. Exposed in Basements)				
	Scope includes, connections of CI pipes with G.I. Puddle flanges provided in retaining wall, including P & F of necessary fittings like thimble pc., tail pc., G.I. flanges, rubber packing, G.I. / S.S. nut-bolt-washer, dripseal joints etc. complete with all respect.				
	Soil line branches shall be connected to main line with using two no. 45 deg. bends. Direct connections with 90 deg. bends are not allowed. (i.e. For all Vertical Soil line connections with Suspended drain lines)				
	Horizontal support spacing for 80mm = 1.0mtr / 100mm = 1.2mtr / 150mm = 1.4mtr / 200mm and higher dia. 1.5mtr. center to center for straight lines.				
	Hanger rod dia. for support clamps – 80 & 100mm = 12mm / 150 & 200mm = 16mm / 300mm = 19mm.				
	(For Suspended drain lines - main & branch at stilt & pump room ceiling piping)				
	Fix Fabricated G.I. Bracket with G.I. 'U' clamps near plumbing shaft (i.e. immediate diversion from vertical line to horizontal line), near bends of main header, near branch tapping with header. (Use 50x50x5 Angle for brackets) and every 5 mtr. Interval for horizontal piping.				
8.1	80mm dia.	RM	3.00	0.00	3.00
8.2	100mm dia. (canopy drain if required).	RM	60.00	0.00	60.00
8.3	150mm dia.	RM	90.00	0.00	90.00
8.4	200mm dia.	RM	0.00	0.00	0.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
9	Providing, fixing, Testing and Commissioning of CI spun 'P' traps for waste line connections with GI brackets and DRIP seal jointing with testing of joints for leak proof. Waste line branches shall be connected to main line with using single 'Y' & 45 deg. bends. Direct connections with 90 deg. bends are not allowed for waste lines.				
	Vertical Waste line connections with Suspended drain lines.				
9.1	80 mm inlet and 80 mm outlet. (for AC drain)	No.	2.00	0.00	2.00
9.2	100 mm inlet and 100 mm outlet. (Toilet / Kit. / Duct / Balcony Waste line)	No.	13.00	0.00	13.00
10	Providing, fixing, Testing and Commissioning of UPVC / PP manually operated backflow preventer (NRV) for rain water lines discharging to main SWD / Nalla from DCB. Including construction of valve chamber of 900 x 600 size as per company specifications and P & F of Heavy duty D.I. cover 104 kg (Item code no = C-250 / MR 17), including excavation in any type of soil . (Chamber / civil items considered separately in respective items)				
	(To be confirm before order)				
10.1	150mm dia.	Nos.	1.00	0.00	1.00
10.2	200mm dia.	Nos.	0.00	0.00	0.00
	TOTAL : "D" CARRIED TO SUMMARY				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
E.	PUMPS AND EQUIPMENTS				
1	Supply, Installing, testing and commissioning of Water Transfer Pump for OH Tank Filling Vertical inline multistage centrifugal pump. Each pump shall have TEFC 2900 RPM, IE3 Rating efficiency, three phase electric motor. The pump shall have cast Iron head & base with SS Shaft/ Impeller/ Intermediate chamber. Pump shall be provided with mechanical seal. The pump shall be suitable for auto/ manual operation.				
	Accessories like NRVs, isolating valves, suction and delivery header / piping, pressure gauge, MS base frame (Synthetic enamel painted over a coat of primer) has to be provided. (Quantities of Valves / pipe / gauges etc. considered in respective BOQ items).				
	Making Suitable cement concrete foundation with cement plaster for pumps with supply & fixing of required Anti vibration rubber pads (150mm x 150mm x 25mm thk. = 4 nos. per pump) and foundation bolts for pumps Bolt Sizes : a) Upto 10 HP Pump - 12mm Ø, b) 10 HP to 50 HP Pump - 16 mm Ø, c) Above 50 HP Pump - 20 mm Ø, d) As per pump manufacturers requirement.				
1.1	Flow rate : 535.33 LPM				
	Head : 103 Mts				
	Purpose : Domestic Water transfer (1 Set = 1W+1SB)	Set	1.00	0.00	1.00
	Pump selected are, Kirlskar make Multistage Vertical inline pump , Model – KCIL32-08-2, 8 stages, 77.73% efficiency, 2900 RPM, Pump coupled with 20 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 32120 LPH / 535.33 LPM / 8.92 LPS. at 103 meter head. OR Lubi make Multistage Vertical inline pump , Model – LCR32-08-2, 8 stages, 75.93% efficiency 2900 RPM, Pump coupled with 20 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 32120 LPH / 535.33 LPM / 8.92 LPS. at 103 meter head. OR Wilo Mather + Platt make Multistage Vertical inline pump , Model – HELIX FIRST V3606/2-5/25/E/KS/415-50, 2900 RPM, Pump coupled with 20 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 32120 LPH / 535.33 LPM / 8.92 LPS. at 103 meter head.				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
1.2	Flow rate : 268 LPM				
	Head : 99 Mts				
	Purpose : Flushing Water transfer (1 Set = 1W+1SB)	Set	1.00	0.00	1.00
	Pump selected are, Kirloskar make Multistage Vertical inline pump , Model – KCIL 15-09, 9 stages, 70.99% efficiency, 2900 RPM, Pump coupled with 10 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 16080 LPH / 268 LPM / 4.46 LPS. at 99 meter head. OR Lubi make Multistage Vertical inline pump , Model – LCR15-09, 9 stages, 71.46% efficiency 2900 RPM, Pump coupled with 10 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 16080 LPH / 268 LPM / 4.46 LPS. at 99 meter head. OR Wilo Mather + Platt make Multistage Vertical inline pump , Model – HELIX FIRST V1610-5/25/E/KS/415-50, 2900 RPM, Pump coupled with 10 HP, 3 Phase, 50 Hz, 415v electric motor(IE3) Having capacity 16080 LPH / 268 LPM / 4.46 LPS. at 99 meter head.				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
2	Supply, Installing, testing and commissioning of Skid mounted terrace booster system with suction & delivery header complete including necessary valves, NRV's, air vessel, electrical panel, dry run protection etc. Each pump shall have TEFC 2900 RPM, IE3 Rating efficiency, three phase electric motor. The pump shall have cast Iron head & base with SS Shaft/ Impeller/ Intermediate chamber. Pump shall be provided with mechanical seal. System shall operate on Auto mode. (other than skid mounted system Valves / pipe / gauges etc. quantities are considered in respective BOQ items).				
	Making Suitable cement concrete foundation with cement plaster for pumps with supply & fixing of required Anti vibration rubber pads (150mm x 150mm x 25mm thk. = 4 nos. per pump) and foundation bolts for pumps Bolt Sizes : a) Upto 10 HP Pump - 12mm Ø, b) 10 HP to 50 HP Pump - 16 mm Ø, c) Above 50 HP Pump - 20 mm Ø, d) As per pump manufacturers requirement.				
	To be order only after confirmation from HOH team.				
2.1	Flow rate : 215 LPM				
	Head : 15 Mts				
	Purpose : Terrace booster System for DOMESTIC (1 Set = 1W+1SB)	Set	0.00	0.00	0.00
	Pump technical details to approve from HOH MEP before order.				
3	Fixing, jointing and testing including labour for fixing in position Irrigation pumps with supply and fixing of required Anti vibration rubber pads (150mm x 150mm x 25mm thk. = 4 nos. per pump) and foundation bolts for pumps Bolt Sizes : a) Upto 10 HP Pump - 12mm Ø, b) 10 HP to 50 HP Pump - 16 mm Ø, c) Above 50 HP Pump - 20 mm Ø including making concrete foundation for pumps. Valves and piping considered in respective BOQ items.				
3.1	Irrigation system pumps (1W+1SB)	Set	1.00	0.00	1.00
	TOTAL : "E" CARRIED TO SUMMARY				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
F.	ELECTRICAL PANELS AND CABLES				
1	Design, fabrication, assembling, wiring, supply, installation, testing and commissioning of following Wall mounted / Floor mounted (including base stand 200 mm height) motor control centre, shall be fabricated out of 14 gauge CRCA sheet steel in form in 3b formation with reinforcement of suitable size angle iron, channel 'T' sections irons and/or flats wherever necessary. Cable gland plates shall be provided on top as well as at the bottom of the panels. The indoor panels shall be IP 42 and outdoor panels shall be IP 54 type. The Panel shall have proper space with required clearance for cables, incoming / outgoing switchgear, CTs, PTs, ammeter, Voltmeter, selector switches, Insulated and properly supported compartmentalized bus bars with heat shrinkable sleeves along with the other control circuit accessories within the panel and any other electrical component mentioned in the Panel SLD.				
	The Panel shall have seven tank pre treatment process comprising of degreasing, rinsing, de-rusting, rinsing, phosphatising, rinsing and passivation followed by powder coat painting having a paint thickness of 60 microns or as specified. The Panel shall be Dust/Vermin proof. Painting as per specifications with 2 coats of red oxide primer and final approved shade of powder coated paint. 2 Nos. earthing terminals shall be provided for 3 phase, 4 wire, 50 Hz supply system. Lifting hooks shall also be provided in case of large panels. as per attached SLD & BOM.				
	Water Level sensor Shall be there in UG and overhead tanks to get Pumps on off Command to the system.				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
1.1	MCC –PLUMBING PANEL (plant room)				
	Incomer				
	160A TP+N MCCB with MP Based OL,SC,ST Releases, Aux. 25KA & RHOM				
	R Y B, Indicating Lamps				
	Off On Trip Indicating Lamp				
	Digital Multi Data Meter with RS 485Port BMS Compatible				
	Current Transformer 150/5A CL.-1 5VA				
	Digital Earth Leakage Relay with CBCT				
	Control MCB 6A TP 10KA C Curve				
	Necessary aux. contact of the I/C MCCB's Off,On,Trip status to BMS				
	Bus Bar				
	160 amps TPN (25KA) aluminium bus bar with heat shrinkable insulation sleeves.				
	Outgoings				
a.	2 Nos. TP MPCB with 20 HP Star delta starter as per type -2 cordination (1 set - 1W+1SB) as outgoing feeder to Domestic transfer pump . Each compartment shall contain CT operated digital electronic ammeter of 0-50 amps range with auto/manual selector switch, on/off push button ,SPP and indicating lamps with MCB's , 'ON/OFF/TRIP' status of motor.				
b.	2 Nos. TP MPCB with 10 HP Star delta starter as per type -2 cordination (1 set - 1W+1SB) as outgoing feeder to Flushing transfer pump . Each compartment shall contain CT operated digital electronic ammeter of 0-50 amps range with auto/manual selector switch, on/off push button ,SPP and indicating lamps with MCB's , 'ON/OFF/TRIP' status of motor.				
c.	(Provision) 2 Nos. TP MPCB with 2.0 HP Direct on line starter as per type -2 cordination (1 set - 1W+1S) as outgoing feeder to Skid mounted terrace Booster System pumps for Domestic . Each compartment shall contain CT operated digital electronic ammeter of 0-10 amps range with auto/manual selector switch, SPP and indicating lamps with MCB's and 'ON/OFF/TRIP' status of motor				

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
	Spare MCCB's of following capacities:				
d.	63 amps TPN MCCB 1 No with TM based OL ,SC Release, Aux 25KA & RHOM with on off indicating lamp with control MCB, 6A SP 10KA C curve.				
e.	40 amps TPN MCB 1 No 10KA C Curve. - 01 nos				
f.	25 amps TPN MCB 1 No 10KA C Curve. - 01 nos				
g.	Necessary cable alleys, internal wiring, and interlocking, earthing for all equipment shall also included.				
	Notes:				
a.	All MCCBs shall be of 25KA breaking capacity and suitable for motor duty application.				
b.	Provision shall be made for providing potential free contacts to all pumps starters for connection to building automation system.				
c.	Provision shall be made for cyclic operation of Domestic & Flushing Pumps in auto mode. i.e. Standby pump shall also operate at an regular interval in auto mode, Sequence timer for pump operation shall be provided.				
d.	Provision shall be made to ensure system protection from dry running of the pump				
e.	All motor MCCB/protecting devices shall be as per TYPE -2 co-ordination.				
f.	Water level controller with sensor,float switch to be provided with Panel.				
	All Panels should be dispatched at site only after inspection at manufacturer workshop by representatives of HOH electrical department. Cost of traveling (i.e.- contractor to make to & fro arrangements for transportation), testing etc. shall be borne by the Contractor. Any Panel dispatched without inspection is bound for rejection.				
	MCC –PLUMBING PANEL AS DESCRIBED ABOVE	Set	1.00	0.00	1.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
3	Supplying & laying of following 1100 volt grade XLPE insulated aluminium / copper conductor armoured cables having outer sheath of fire retardent extruded (FRLS) compound as per specification in existing trenches, cable trays, ducts over bed of sand, clamped to wall with suitable clamps, saddles fixing bolts including connecting testing and commissioning. All cables to be labelled (black anodized) at both the ends.				
3.1	3c x 10 sq. mm copper armoured cable .	RM	100.00	0.00	100.00
3.2	3c x 4 sq. mm copper armoured cable .	RM	150.00	0.00	150.00
3.3	3c x 6 sq. mm copper armoured cable .	RM	100.00	0.00	100.00
3.4	3c x 2.5 sq. mm copper armoured cable .	RM	0.00	0.00	0.00
3.5	4c x 2.5 sqmm copper armoured cable.	RM	0.00	0.00	0.00
3.6	4c x 4 sqmm copper armoured cable.	RM	200.00	0.00	200.00
4	Supply & fixing of Cable end termination of the following XLPE insulated aluminium/copper conductor armoured cables of 1100 volt grade including supplying and fixing of bimettalic crimping lugs, double compression glands with earthing facility and cable sockets etc. complete as required.				
4.1	3c x 10 sq. mm copper armoured cable .	Nos.	8.00	0.00	8.00
4.2	3c x 4 sq. mm copper armoured cable .	Nos.	4.00	0.00	4.00
4.3	3c x 6 sq. mm copper armoured cable .	Nos.	8.00	0.00	8.00
4.4	3c x 2.5 sq. mm copper armoured cable .	Nos.	0.00	0.00	0.00
4.5	4c x 2.5 sqmm copper armoured cable.	Nos.	0.00	0.00	0.00
4.6	4c x 4sqmm copper armoured cable.	Nos.	2.00	0.00	2.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
5	Supply and fixing of following Earthing , bare GI/copper tapes / wires including all necessary fixing accessories and effecting connections as per specifications.				
5.1	25 x 3 mm. GI. Strip.	RM	160.00	0.00	160.00
5.2	10 Gauge Bare GI. Wire.	RM	100.00	0.00	100.00
5.3	08 Gauge Bare GI. Wire.	RM	160.00	0.00	160.00
6	Design, Fabrication, Supply, Unloading, Handling, Installation, Commissioning of Perforated ladder type Cable Trays of Pre galvanised 02 mm thick : As per IS: 2062 /1079 & Galvanising : As per IS: 2629/4759 ,all accessories include. Ladder type cable trays of following sizes, suspended from ceiling or fixed vertically on wall / column as per route directed at site, as approved complete as per technical specifications, drawings & relevant standards & site conditions.				
6.1	size : 150mm wide.	RM	60.00	0.00	60.00
6.2	size : 300mm wide.	RM	30.00	0.00	30.00
6.3	size : 450mm wide.	RM	0.00	0.00	0.00
6.4	100 x 50 mm GI trunking	RM	20.00	0.00	20.00
TOTAL : "F" CARRIED TO SUMMARY					

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
G.	MISCELLANEOUS ITEMS				
1	Providing and fixing G.I. blank flanges, making flanged joints with G.I. bolts, G.I. nuts, G.I. washers and rubber insertion 3 mm thick, testing the flange and joints etc. complete for pipes of following diameter. (for unused / spare plumbing sleeves of UG Tanks & Retaining wall)				
1.1	80 mm dia	Nos.	1.00	0.00	1.00
1.2	100 mm dia	Nos.	1.00	0.00	1.00
1.3	150 mm dia	Nos.	1.00	0.00	1.00
2	Fabrication, Supply, installation & fixing of puddle flanges, 600mm long for the storage tanks. The puddle flanges shall be fabricated out of MS pipes and plates as per the drawing enclosed and hot dipped galvanized before installations for O.H. Tanks including Fire Fighting Tanks.				
2.1	50 mm dia (Screwed ends).	Nos.	17.00	0.00	17.00
2.2	80 mm dia (Flanged ends).	Nos.	2.00	0.00	2.00
2.3	100 mm dia (Flanged ends).	Nos.	8.00	0.00	8.00
2.4	150 mm dia (Flanged ends).	Nos.	0.00	0.00	0.00
3	Scaffolding for plumbing works in shaft (if required) with all safety measures for labourers working on scaffolding.	Sft.	200.00	0.00	200.00

OAKWOOD BUILDING AT KALYAN

Item	Description	Unit	Quantity		Total
			Tower	Non Tower	
4	Pipe label works for all plumbing pipes (All Water supply lines - Flushing / Domestic / Solar etc. All Drainage lines - Soil / Waste / Vent / Rain water etc. All types of pump delivery lines and pumps, All UG & OH Tanks etc.) Labels should be positioned on the pipes so they can be easily read. Proper label placement is on the lower side of the pipe if the employee has to look up to the pipe, on the upper side of the pipe if the employee has to look down towards the pipe, or directly facing the employee if on the same level as the pipe. Labels should be located near valves, branches, where a change in direction occurs, on entry/re-entry points through walls or floors, and on straight segments with spacing between labels that allows for easy identification and as per requirement of engineer incharge.				
	For Label work with arrow marks – “Flushing tank”, “Domestic tank”, “Raw Water tank” & “Fire tank” etc. shall be marked on each cover respectively, after enamel paint, including label works for various areas as per pictures given in specifications, terrace - podium main loop & branch, down takes - uptake main & branch, pump delivery lines, pumps, sump pump lines, tank overflow / cleanouts, suspended CI drain lines, Suspended pvc drain or rain water pipes etc.				
4.1	All plumbing pipe label works as per above details as well as details and pictures shown in specifications. (TA+NTA)	Job	1.00	0.00	1.00
	All Toilets - Flushing / Domestic branch and All Kitchens - Domestic branch = 25%.				
	Terrace loop & branch pipes, All OH Tank capacity details, names on cover, tank overflow / cleanouts etc. = 15%.				
	Podium loop & branch pipes (where applicable), All UG Tank capacity details, names on cover, tank overflow / cleanouts etc. = 15%				
	Domestic, Flushing, Sump pump, other pump details, all pump delivery lines = 15%				
	Suspended CI Main & Branch drain lines = 15%				
	Suspended PVC Main & Branch drain lines (terrace / surface drain / garden drain) = 15%				
5	Supply and handover of Manual (Hand) Drain Cleaning Tool with steel coil.				
	Material to be handover to HOH Plumbing department with necessary test and warrantee / guarantee certificates of manufacturer.				
5.1	with 12 feet coil (Varie)	Nos.	2.00	0.00	2.00
5.2	with 25 feet coil (Varie)	Nos.	2.00	0.00	2.00
TOTAL : "G" CARRIED TO SUMMARY					